

# Shunting Inhibition and Dendritic Branching Shape Local Credit Assignment

Houman Safaai<sup>1</sup>   Maceo Richards<sup>1</sup>  
Bernardo L. Sabatini<sup>1,2</sup>

<sup>1</sup> Kempner Institute for the Study of Natural and Artificial Intelligence  
at Harvard University

<sup>2</sup> Department of Neurobiology, Howard Hughes Medical Institute,  
Harvard Medical School, Boston, MA 02115, USA

June 26, 2026

## Abstract

Biological neurons assign credit across branching dendrites, where synaptic drive, reversal potential, local voltage, and input resistance jointly determine plasticity. We study conductance-based dendritic networks with E/I synapse banks, shunting inhibition, and tree-structured branch-to-soma coupling, and ask when restricted somatic feedback can approximate compartment-specific backpropagated errors. Exact gradients factor into *local eligibility*  $\times$  *compartment error*: the eligibility uses presynaptic activity, driving force, and input resistance, while the fast non-local term is a path-specific error obtained by transporting a soma error through dendritic gains. This factorization turns local learning into a credit-signal compression problem. We test the hypothesis that shunting inhibition helps when it reshapes the compartment-error field to better match global-scalar, per-soma, low-rank, or path-structured feedback. Exact-gradient reconstruction verifies the factorization; path-gain, rank, broadcast-fidelity, inhibition-intervention, and transported-error-oracle diagnostics support the proposed mechanism. Under nonnegative conductances and per-soma 5F feedback, shunting LocalCA remains 5–6 percentage points below matched backpropagation on MNIST, Fashion-MNIST, and figure-ground MNIST, indicating that feedback-field fidelity remains a major bottleneck. These results show how E/I conductance, shunting inhibition, and dendritic branching can reshape credit-signal geometry in restricted local learning.

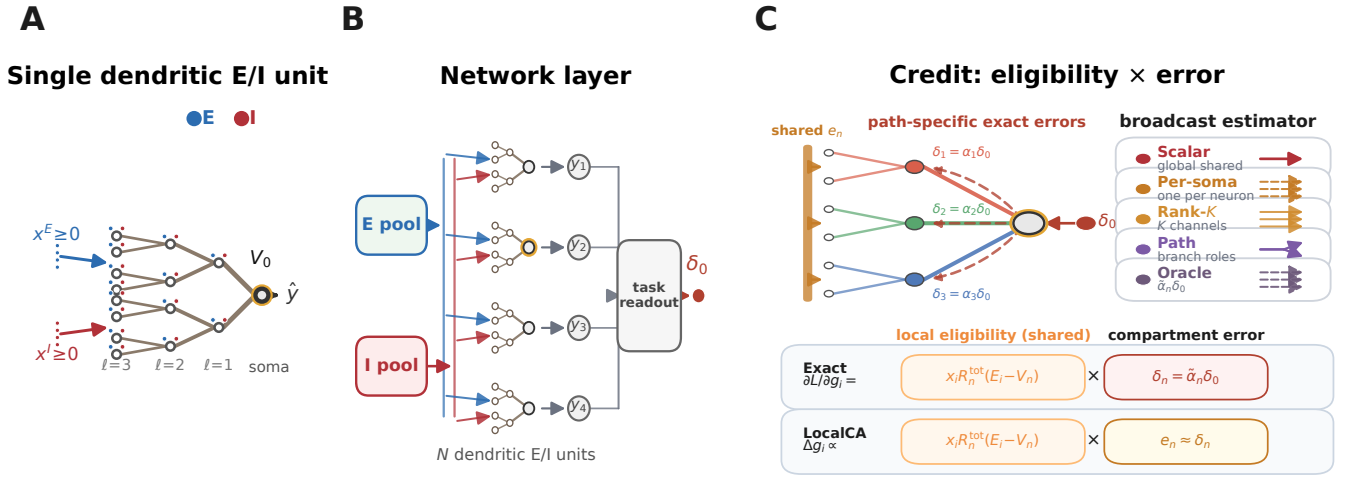
## 1 Introduction

Unlike point units in standard neural networks, biological neurons assign credit across branching dendritic trees. A synapse receives more than presynaptic activity and a scalar postsynaptic activation: it also has access to local membrane voltage, synaptic reversal potential, total conductance, and input resistance. Inhibitory synapses can therefore do more than subtract activity: by increasing total local conductance, inhibition can shunt voltage and change the gain of every dendritic path passing through that compartment.

We examine whether these biophysical ingredients can facilitate local credit assignment (LocalCA). Backpropagation [16] would assign a distinct error to every dendritic compartment, whereas a biologically plausible supervisory signal is more likely to arrive as a low-bandwidth somatic or modulatory broadcast [21]. The central question is therefore not whether such a broadcast can reproduce unconstrained backpropagation in every setting, but when the exact dendritic error field is simple enough for it to approximate. Our contribution is not simply to add dendritic structure to a neural network, but to show how conductance-based synapses, shunting inhibition, and dendritic topology change the geometry of the credit signal itself.

Starting from conductance-based dendritic voltage equations [1], we derive exact gradients for dendritic trees (Theorem 1). Each synaptic gradient factorizes into a synapse-local eligibility term and a single non-local compartment error:

$$\text{gradient} = \underbrace{\text{local eligibility}}_{\text{presynaptic drive, } E-V, R^{\text{tot}}} \times \underbrace{\text{compartment error}}_{\text{path-specific non-local term}} .$$



**Figure 1: Model and credit assignment.** (A) Single dendritic E/I unit with nonnegative excitatory and inhibitory input streams, branch synapse banks, and a gold-ringed soma matching the highlighted unit in B. (B) Layer of dendritic E/I units projecting to a task readout;  $\delta_0 = \partial L / \partial V_0$  is the layer-wide soma/core error from which feedback is constructed. (C) Credit factorization: exact updates use the same local eligibility multiplied by path-specific compartment errors  $\delta_{u,n}$ , whereas LocalCA keeps the eligibility and replaces  $\delta_{u,n}$  with a broadcast estimator  $e_{u,n}$ ; global scalar, per-soma, Rank-K, Path, and Oracle broadcasts trade bandwidth for fidelity.

The compartment error is path-specific: under identity upward transfer it is the corresponding soma error multiplied by a conductance-stage path gain through that dendritic tree. LocalCA preserves the exact local eligibility and replaces this path-specific error with a broadcast field.

**What is local in LocalCA?** The fast eligibility variables are synapse- or compartment-local: presynaptic drive, local voltage, driving force, and input resistance. The 4F and 5F variants additionally use slowly estimated branch-level statistics. The fast task-dependent non-local quantity is the compartment error, approximated by a broadcast field  $e_n$ . Thus LocalCA variants differ by the broadcast estimator and optional branch-level modulators, while preserving the same local eligibility mechanism.

This factorization turns local credit assignment into a feedback-compatibility problem. Intrinsic low rank of the exact compartment-error matrix is useful, but it is not sufficient: the error field must also align with the particular feedback object available to the rule. We therefore distinguish global scalar broadcast, neuron-wise/per-soma broadcast, random low-rank feedback, path-structured feedback, and exact transported-error oracles. Dendritic branching creates the paths; E/I synapse banks set the local conductance state; and shunting inhibition changes the input resistance along each path. Shunting is not assumed to help monotonically. It helps restricted feedback when inhibitory conductance attenuates high-gain paths or otherwise makes the exact error field better aligned with the available broadcast.

**Related work.** This work builds on models of dendritic signal integration, branch-level nonlinear computation, divisive normalization, three-factor plasticity, and biologically plausible teaching signals [35, 1, 3, 4, 36, 9, 12, 8, 18, 19, 20, 27, 28, 29, 23, 46]. It also connects to machine-learning approaches that replace exact backpropagation with local, random, predictive, equilibrium, perturbation, or broadcast feedback signals [16, 15, 14, 17, 32, 22, 33, 34, 30, 31, 37, 38, 39, 40, 41]. Our contribution is complementary: we derive the exact conductance-tree credit object and test when shunting inhibition and dendritic morphology make that object broadcast-compatible.

We test this chain directly. Exact-gradient reconstruction verifies the factorization numerically. Gradient-fidelity diagnostics are used as diagnostics, not as a claim of backpropagation equivalence: shunting improves directional alignment relative to an additive control, but the absolute cosine remains modest, while scale mismatch is substantially reduced. Exact-error rank and direct broadcast-fidelity diagnostics show that shunting can make the compartment-error field easier for the feedback field used by the learning rule to approximate. Transported-error oracle experiments show that supplying the exact path-transported compartment error substantially reduces the local-learning gap. Together,

these results identify feedback fidelity as an important bottleneck in these experiments.

**Contributions.** This paper makes three contributions. First, it derives the conductance-stage path gain, its local-eligibility and path-error factorization, and a closed-form inhibitory path-gain ratio. Second, it identifies path-gain and broadcast-compatibility metrics that mediate whether restricted somatic feedback can approximate exact compartment errors. Third, it shows empirically that shunting inhibition can improve this approximation while also identifying regimes where scalar or low-rank feedback fails.

**Scope of evidence.** Analytical statements concern the conductance-tree factorization, path gains, and inhibitory path-gain derivatives under the stated steady-state assumptions. Simulation controls test exact reconstruction, broadcast fidelity, transported-oracle upper bounds, and additive/shunting comparisons. Biological interpretations are restricted to branch-local inhibition and morphology-dependent credit routing; transported-error experiments are oracle diagnostics, not proposed biological learning rules.

## 2 Compartmental Voltage Model and Gradient Derivation

For each dendritic compartment, we use a steady-state conductance model derived from discretized passive cable dynamics [1, 2]. In normalized units, leak and inhibitory reversal potentials are set to 0, the excitatory reversal potential is set to 1, and leak conductance is fixed at 1 (i.e., all conductances are expressed relative to leak). Each compartment computes a conductance-weighted voltage: excitatory conductance pulls the voltage toward the excitatory reversal potential, leak and shunting inhibition pull it toward zero, and dendritic coupling transfers child voltages upward toward the soma. This form makes local sensitivity depend on both driving force ( $E - V$ ) and total input resistance  $R^{\text{tot}}$ . Throughout the paper we write dendritic morphologies as rooted trees  $[b_1, b_2, \dots, b_D]$ , where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. Thus  $[3, 3]$  means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches, for 9 distal leaves per soma. In the main feedforward experiments, synaptic inputs terminate on dendritic branches rather than directly on the soma. Each branch layer carries explicit excitatory conductance-based synapse banks and, when enabled, learned inhibitory branch-level conductance-based synapse banks driven by nonnegative inputs. This is input-driven shunting inhibition, not a recurrent lateral inhibitory circuit; explicit inhibitory-cell controls are reported in the appendix. The additive and shunting cores are architecture-matched at the tree level and differ only in the branch-level synaptic integration rule. These ingredients play distinct roles in the credit-assignment argument. Branching creates the path structure. E/I synapse banks determine the local conductance state. Shunting turns inhibitory conductance into a multiplicative change in path gain through input resistance. The local learning rule then asks whether a low-bandwidth teaching signal can stand in for the exact path-specific error induced by that tree. **Notation.** We use  $\delta_{0,u}^{(\ell),V} = \partial L / \partial V_{0,u}^{(\ell)}$  for the exact voltage-space error of soma  $u$  in layer  $\ell$ ,  $\delta_{0,u}^{(\ell),a} = \partial L / \partial a_{0,u}^{(\ell)}$  for the corresponding activation-space error when a soma activity  $a_0 = f_0(V_0)$  is used,  $\boldsymbol{\delta}_0$  for the layer-wide soma/core teaching vector used operationally,  $\bar{\delta}$  for the global scalar obtained by averaging soma coordinates within an example, and  $\delta_{u,n}^{(\ell)} = \partial L / \partial V_{u,n}^{(\ell)}$  for compartment  $n$  of neuron  $u$  in layer  $\ell$ . We use  $\alpha_n^{\text{cond}}$  for the conductance-stage path gain from compartment  $n$  to the soma and  $\tilde{\alpha}_n$  for the effective path gain after including any post-voltage activation derivatives; when upward transfer is identity,  $\tilde{\alpha}_n = \alpha_n^{\text{cond}}$ . We use  $N_E$  and  $N_I$  for the number of excitatory and inhibitory synapses per branch,  $r_n^{4F}$  for the empirical 4F branch-soma covariance proxy,  $\phi_n$  for the 5F bounded branch-level preconditioner,  $\lambda_{\text{mix}}$  for the pathway-broadcast mixing weight, and  $\kappa_j$  for depth-dependent modulation in morphology extensions. All conductances and presynaptic activities in the main experimental setting are nonnegative; additive voltages may be signed comparators, but the conductance variables that define the shunting model are nonnegative.

*Voltage equation and local sensitivities.* Consider compartment  $n$  with synaptic inputs  $j$  (activity  $x_j$ , reversal  $E_j$ , conductance  $g_j^{\text{syn}} \geq 0$ ) and dendritic inputs from children (voltage  $V_j$ , conductance  $g_j^{\text{den}} \geq 0$ ). The steady-state voltage is:

$$V_n = \frac{\sum_j E_j x_j g_j^{\text{syn}} + \sum_j V_j g_j^{\text{den}}}{\underbrace{\sum_j x_j g_j^{\text{syn}} + \sum_j g_j^{\text{den}} + 1}_{g_n^{\text{tot}}}}, \quad R_n^{\text{tot}} = 1 / g_n^{\text{tot}}. \quad (1)$$

$V_n$  is a convex combination of reversal potentials, child voltages, and leak. Writing  $\mathcal{S}_n = \{0\} \cup \{E_j\}_j \cup \{V_j\}_j$ , we have  $\min \mathcal{S}_n \leq V_n \leq \max \mathcal{S}_n$  and  $0 < R_n^{\text{tot}} \leq 1$ . The local sensitivities follow directly:

**Proposition 1** (Local Sensitivities).

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = x_i R_n^{\text{tot}} (E_i - V_n), \quad \frac{\partial V_n}{\partial V_i} = g_i^{\text{den}} R_n^{\text{tot}}, \quad \frac{\partial V_n}{\partial g_i^{\text{den}}} = R_n^{\text{tot}} (V_i - V_n). \quad (2)$$

Eq. (2) gives the eligibility factors used by the local rules below: presynaptic activity or voltage difference, driving force, and input resistance. If a child transmits post-voltage activity  $a_i = f_i(V_i)$  rather than raw voltage, the dendritic-conductance derivative is  $R_n^{\text{tot}}(a_i - V_n)$  and the child-voltage error includes the local factor  $f'_i(V_i)$ .

*Shunting inhibition as divisive gain control.* An inhibitory synapse with  $E_{\text{inh}} \approx 0$  contributes current  $(0 - V_n)x_j g_j^{\text{syn}}$  and increases  $g_n^{\text{tot}}$ . Its sensitivity is  $\partial V_n / \partial g_j^{\text{syn}} = -x_j R_n^{\text{tot}} V_n$ , which corresponds to multiplicative attenuation (divisive normalization). Shunting is divisive at the voltage level, but its effect on firing rates can be subtractive in some regimes [10]. Inhibitory plasticity can balance excitation dynamically [11]; our learned inhibitory conductances provide a trainable balancing mechanism. The same denominator also gives the direct connection to path gain. Writing  $G_n^E(x)$  and  $G_n^I(x)$  for the total excitatory and inhibitory synaptic conductance impinging on compartment  $n$ ,

$$R_n^{\text{tot}}(x) = (1 + G_n^E(x) + G_n^I(x) + G_n^{\text{den}})^{-1}.$$

Increasing  $G_n^I(x)$  lowers  $R_n^{\text{tot}}(x)$  and therefore multiplicatively suppresses every upstream path whose error must pass through compartment  $n$ . In this sense inhibition can gate credit flow by changing the path gain, even when the inhibitory drive is feedforward rather than lateral. Prop. 2 formalizes this statement after the tree path gain is defined.

*Exact gradients for dendritic trees.* For one neuron/tree in layer  $\ell$ , let  $V_{0,u}^{(\ell)}$  be the somatic voltage and define the exact voltage-space soma error as  $\delta_{0,u}^{(\ell),V} := \partial L / \partial V_{0,u}^{(\ell)}$ . If a decoder receives soma activity  $a_0 = f_0(V_0)$  rather than raw voltage, the decoder supplies an activation-space error  $\delta_0^a$  and the voltage-space boundary is  $\delta_0^V = f'_0(V_0) \delta_0^a$ . The theoretical derivation is stated for the conductance-stage voltage before any post-voltage reactivation. When a branch applies a monotone reactivation before transmitting activity upward, the same recursion holds with additional local derivative factors along the path. All exact-error diagnostics below are computed as pre-activation voltage errors, including activation-derivative factors when reactivation is enabled.

**Theorem 1** (Conductance-Stage Backpropagation on a Dendritic Tree). *For a rooted dendritic tree with soma at node 0 and unique parent  $p(n)$  for each non-somatic compartment, suppose the parent receives either the pre-activation voltage  $V_n$  or a local post-voltage activity  $a_n = f_n(V_n)$ . For identity upward transfer, the loss gradient satisfies*

$$\frac{\partial L}{\partial V_n} = \frac{\partial L}{\partial V_{p(n)}} R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}, \quad (3)$$

with boundary condition  $\partial L / \partial V_0 = \delta_0^V$ . With post-voltage transfer  $a_n = f_n(V_n)$ , the local factor is the product of  $f'_n(V_n)$ ,  $R_{p(n)}^{\text{tot}}$ , and  $g_{n \rightarrow p(n)}^{\text{den}}$ . Because the graph is a tree, each compartment has a unique path to the soma. Defining the conductance-stage path gain

$$\alpha_n^{\text{cond}} = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \alpha_n^{\text{cond}} \delta_0^V, \quad (4)$$

with  $\alpha_0^{\text{cond}} = 1$  under identity transfer. In the post-voltage case, the exact effective path gain is

$$\tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \tilde{\alpha}_n \delta_0^V, \quad (5)$$

where  $f'_i(V_i)$  is omitted for identity transfer.

*Proof.* Apply the chain rule on the tree-structured computation graph using Prop. 1.  $\square$

The one-step recursion in Eq. (3) and the path product in Eq. (4) separate local conductance transfer from the non-local somatic error for one soma/tree and one layer. Supplying an approximate layer-soma teaching vector, reusing a decoder-derived vector at another layer, or compressing  $\delta_{0,u}^{(\ell),V}$  into a per-soma or global scalar broadcast is an additional feedback-approximation step introduced by LocalCA, not a consequence of the theorem itself.

**Proposition 2** (Inhibitory control of conductance-stage path gain). *Let  $\mathcal{A}(n)$  be the set of compartments on the path from compartment  $n$  to the soma, excluding  $n$  and including the parent compartments through which the error must pass. Holding dendritic coupling conductances fixed, an added inhibitory conductance  $\Delta G_k^I(x) \geq 0$  at any  $k \in \mathcal{A}(n)$  changes the conductance-stage path gain by*

$$\frac{\alpha_n^{\text{cond}}(x; \Delta G^I)}{\alpha_n^{\text{cond}}(x; 0)} = \prod_{k \in \mathcal{A}(n)} \frac{g_k^{\text{tot}}(x)}{g_k^{\text{tot}}(x) + \Delta G_k^I(x)}, \quad \frac{\partial \log \alpha_n^{\text{cond}}}{\partial G_k^I} = \begin{cases} -R_k^{\text{tot}}, & k \in \mathcal{A}(n), \\ 0, & k \notin \mathcal{A}(n). \end{cases} \quad (6)$$

*Proof.* Substitute  $R_k^{\text{tot}} = 1/g_k^{\text{tot}}$  into the path-gain product in Eq. (4). Adding inhibitory conductance changes only the denominator at compartments on the path from  $n$  to the soma, which gives the product ratio and the log derivative in Eq. (6).  $\square$

Prop. 2 makes two directed predictions. First, inhibition is a path gate: it attenuates all credit paths below the inhibited compartment. Second, inhibition is useful for restricted-feedback local learning only when this attenuation makes the distribution of  $\alpha_n^{\text{cond}}$  more uniform or better aligned with the task. If inhibition is too strong, too heterogeneous, or generated by a poorly calibrated inhibitory population, it can suppress useful signals and hurt learning. In the additive control the same inhibitory input changes voltages and local eligibilities, but it does not enter the conductance-stage path recursion as a multiplicative resistance gate.

**When does shunting compress path gains?** The attenuation identity above is not by itself a concentration result. Let  $z_n = \log \alpha_n^{\text{cond}}$  be the log conductance-stage path gain for compartment  $n$ . If added inhibitory conductance contributes a path-dependent attenuation

$$\beta_n(x) = \sum_{k \in \mathcal{A}(n)} \log \left( 1 + \frac{\Delta G_k^I(x)}{g_k^{\text{tot}}(x)} \right), \quad z'_n = z_n - \beta_n, \quad (7)$$

then, over compartments or examples,

$$\text{Var}(z') = \text{Var}(z) + \text{Var}(\beta) - 2\text{Cov}(z, \beta). \quad (8)$$

Thus shunting narrows the log path-gain field exactly when

$$\text{Cov}(z, \beta) > \frac{1}{2} \text{Var}(\beta). \quad (9)$$

Eq. (7) follows by taking the logarithm of the product ratio in Prop. 2; Eq. (8) then applies the variance identity for  $z - \beta$ . Eq. (9) is a diagnostic condition under one averaging measure, not the empirical mechanism claimed below. In finite trained networks, the covariance margin can disagree with CV and exact-error-rank summaries because these statistics average over different compartment and sample axes. We therefore use direct path-gain dispersion, exact-error compressibility, and inhibition-intervention measurements as the empirical tests, and report the covariance margin as a boundary check in the appendix.

**Corollary 1** (Local–Global Factorization). *The exact synaptic gradient at compartment  $n$  factorizes as:*

$$\frac{\partial L}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\text{synapse-local eligibility}} \cdot \underbrace{\frac{\partial L}{\partial V_n}}_{\text{compartment error}}, \quad (10)$$

where the eligibility term depends only on quantities available at synapse  $i$  (presynaptic activity  $x_i$ , input resistance  $R_n^{\text{tot}}$ , driving force  $E_i - V_n$ ), and the compartment error  $\partial L / \partial V_n$  is the sole non-local quantity.

**Proposition 3** (Path-gain dispersion controls 3F alignment). *For a single example before batch averaging, consider one soma/tree with the exact voltage-space soma error  $\delta_0^V$  supplied, and write the local eligibility for conductance parameter  $i$  as  $z_i$ . Let  $n(i)$  be the compartment containing parameter  $i$ . Under identity transfer replace  $\tilde{\alpha}_{n(i)}$  below by  $\alpha_{n(i)}^{\text{cond}}$ ; with post-voltage transfer use the effective gain  $\tilde{\alpha}_{n(i)}$ . Assume both gradient vectors are nonzero, equivalently  $\sum_i w_i > 0$  and  $\sum_i w_i \tilde{\alpha}_{n(i)}^2 > 0$ . The exact and 3F per-soma gradients are*

$$g_i^{\text{BP}} = z_i \tilde{\alpha}_{n(i)} \delta_0^V, \quad \hat{g}_i^{\text{3F}} = z_i \delta_0^V. \quad (11)$$

*If transfer derivatives and conductance couplings are nonnegative,  $\tilde{\alpha}_{n(i)} \geq 0$ , so corresponding nonzero components have the same sign. Their cosine is*

$$\cos(\hat{g}^{\text{3F}}, g^{\text{BP}}) = \frac{\sum_i w_i \tilde{\alpha}_{n(i)}}{\sqrt{\sum_i w_i} \sqrt{\sum_i w_i \tilde{\alpha}_{n(i)}^2}} = \frac{1}{\sqrt{1 + \text{CV}_w(\tilde{\alpha}_{n(i)})^2}}, \quad w_i = (z_i \delta_0^V)^2. \quad (12)$$

*Proof.* Substitute Eq. (11) into the cosine formula and cancel the common signed eligibility factors through  $w_i$ . The final equality is the weighted identity  $E_w[\tilde{\alpha}^2] = E_w[\tilde{\alpha}]^2(1 + \text{CV}_w(\tilde{\alpha})^2)$ .  $\square$

Prop. 3 is conditional on the exact layer-soma error. It explains why low effective path-gain dispersion is a useful within-tree credit geometry, while also making clear that practical multi-layer LocalCA must approximate the layer-soma error before this within-tree argument applies. The factorization in Eq. (10) isolates the fast task-error-dependent non-local quantity as  $\partial L/\partial V_n = \alpha_n^{\text{cond}} \delta_0^V$  under identity upward transfer and  $\partial L/\partial V_n = \tilde{\alpha}_n \delta_0^V$  when post-voltage transfer is enabled. The implemented updates in Eqs. (13)–(14) are batch averages of products, so Prop. 3 should be read as the pre-batch geometry of per-example gradient contributions. The 4F and 5F factors introduced below use slower voltage statistics, so they are empirical reliability preconditioners rather than additional error signals. Any practical local rule therefore depends on how well its broadcast signal approximates that exact compartment error, which we evaluate in the gradient-fidelity diagnostic below. In the models studied here, shunting improves this approximation by reducing and stabilizing  $R_n^{\text{tot}}$ , which narrows the spread of local sensitivities across the tree. In the empirical networks, each dendritic branch layer may optionally apply a post-voltage reactivation  $f(V_n)$  before passing activity to the next stage; unless otherwise noted, the manuscript sweeps use a learnable bounded tanh reactivation.

### 3 Local Learning Rules

*Broadcast error approximation.* We approximate the exact compartment error  $\partial L/\partial V_n$  (Corollary 1) with a broadcast signal  $e_n^V$  derived from a soma/core teaching vector. For a linear readout this vector is  $W_{\text{dec}}^\top (\partial L/\partial \hat{y})$ ; for a nonlinear decoder we use the corresponding decoder-input Jacobian product  $J_{\text{dec}}(h_{\text{core}})^\top (\partial L/\partial \hat{y})$  so that feedback is defined in soma/core coordinates. If this vector is used at an earlier matched-width dendritic layer, practical LocalCA assumes neuron-index correspondence and reuses the coordinate as an approximate layer-soma error. This inter-layer teaching approximation is distinct from the within-tree question addressed by Theorem 1: given a layer-soma error, how faithfully can a restricted branch-level field approximate  $\partial L/\partial V_n$ ? The main competence and mechanism runs use per-soma feedback: when soma/core dimension matches a layer’s neuron dimension, each neuron receives one soma-specific teaching coordinate and the same value is shared across that neuron’s dendritic compartments. Conditional on an exact layer-soma error, per-soma sharing isolates the within-tree approximation. Practical multi-layer LocalCA additionally approximates earlier-layer soma errors by reusing final-core coordinates at matched-width layers, as separated explicitly in Fig. 2D. A stricter scalar mode compresses the soma/core vector to one number per example,  $\tilde{\delta}_b = d_0^{-1} \sum_{c=1}^{d_0} \delta_{0,bc}^a$ , and broadcasts that number to all neurons and compartments. Thus scalar feedback also discards heterogeneity across soma-error coordinates. For higher-bandwidth controls,  $P_K$  is a fixed random projection from soma-error coordinates to  $K$  feedback channels,  $q_{n,k}$  are fixed compartment-specific mixing coefficients, and  $\Gamma_n$  denotes the pathway-structured transport map for compartment  $n$ . We distinguish five feedback objects: **global scalar broadcast**,  $e_n = \tilde{\delta} \mathbf{1}$ ; **per-soma shared broadcast**,  $e_{u,n}$  is one coordinate shared across the compartments of neuron  $u$ ; **rank- $K$  random broadcast**,  $c = P_K \delta_0^a$  and  $e_n = \sum_k q_{n,k} c_k$ ; **path-structured broadcast**,

$e_n = \lambda_{\text{mix}} \bar{\delta} \mathbf{1} + (1 - \lambda_{\text{mix}}) \Gamma_n(\delta_0^a)$  with pathway roles motivated by vectorized dendritic teaching signals [45]; and the **transported oracle**,  $e_n^{\text{transport}} = \tilde{\alpha}_n \delta_0^V = \partial L / \partial V_n$ , an analysis upper bound because it supplies the exact path-transported compartment error from Theorem 1. Operationally, path-transport runs with reactivation enabled transport activation-space errors through parent reactivation slopes and multiply by the local  $f'_n(V_n)$  before conductance updates; with identity transfer this reduces to  $\alpha_n^{\text{cond}} \delta_0^V$ . Thus the main feedforward result tests neuron-wise/per-soma feedback rather than a single layer-wide scalar. Under shunting, this restricted field supports nontrivial learning on the standard classification tasks, while scalar, routed, and rank-bridge controls in the appendix expose regimes where lower-bandwidth feedback collapses branch identity and richer feedback is needed. A local-mismatch heuristic is included only as an appendix control because it performs much worse than the somatic-error broadcasts (Appendix A).

*Three-factor learning rule (3F).*

**Definition 1** (3F Update). *For synaptic and dendritic conductances, with  $a_j = V_j$  under identity upward transfer and  $a_j = f_j(V_j)$  when post-voltage transfer is enabled:*

$$\widehat{\nabla}_{g_j^{\text{syn}}} L = \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n^V \rangle_B, \quad \widehat{\nabla}_{g_j^{\text{den}}} L = \langle R_n^{\text{tot}} (a_j - V_n) e_n^V \rangle_B, \quad (13)$$

where  $\langle \cdot \rangle_B$  denotes the batch average.

The three factors are presynaptic activity  $x_j$  (or a transmitted child activity difference), postsynaptic modulation through driving force and input resistance, and the broadcast voltage-space error  $e_n^V$ . The same rule applies to excitatory and inhibitory synapses. The sign difference comes from the driving force  $(E_j - V_n)$ . We write  $\widehat{\nabla}_g L$  as the local gradient estimate supplied to the optimizer; the optimizer applies the usual learning rate and descent step. Equivalently, one could absorb the sign convention into  $e_n^V$  and treat it as a descent signal.

**Additive control.** Rule (13) is the local gradient of the *shunting* voltage  $V_n = \sum E_j g_j x_j / g_n^{\text{tot}}$ . For the additive E/I control, inhibition enters as a fixed signed voltage contribution rather than through the conductance denominator:

$$V_n^{\text{add}} = \sum_{j \in E} g_j^E x_j^E - \sum_{j \in I} g_j^I x_j^I + \sum_k g_k^{\text{den}} a_k.$$

Equivalently,  $\partial V_n^{\text{add}} / \partial g_j^{\text{syn}} = s_j x_j$  with  $s_j = +1$  for excitatory and  $s_j = -1$  for inhibitory synapses, and  $\partial V_n^{\text{add}} / \partial g_k^{\text{den}} = a_k$ . The additive local gradient therefore has no driving-force or  $R_n^{\text{tot}}$  terms ( $R_n^{\text{tot}} \equiv 1$  by definition since there is no denominator). For the identity-transfer additive tree, the exact upward gain is  $\alpha_n^{\text{add}} = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} g_{i \rightarrow k}^{\text{den}}$ ; with reactivation,  $\tilde{\alpha}_n^{\text{add}} = \prod_{(i \rightarrow k)} f'_i(V_i) g_{i \rightarrow k}^{\text{den}}$ . Thus additive controls have an exact tree gain, but inhibition does not enter that gain through a conductance denominator. Throughout, each architecture uses the learning rule derived from its own forward-pass dynamics. This keeps the comparison architecture-matched rather than applying a shunting-derived rule to an additive model. In implementations with unconstrained raw parameters  $\theta$  and  $g = \text{softplus}(\theta)$ , Eqs. (13)–(14) give conductance-space gradients; before assigning gradients to  $\theta$ , we multiply by  $\partial g / \partial \theta$ . The exact-gradient reconstruction diagnostics include this parameterization factor. The compact mechanism table below summarizes the matched shunting/additive comparison.

| Property              | Shunting / conductance core                               | Additive control                                      |
|-----------------------|-----------------------------------------------------------|-------------------------------------------------------|
| Voltage               | Normalized by total conductance                           | Signed E/I voltage sum                                |
| Inhibition            | Divisive conductance load                                 | Subtractive signed term                               |
| Synaptic eligibility  | $x R^{\text{tot}} (E - V) e$                              | $s x e$ , $s \in \{+1, -1\}$                          |
| Driving force         | Present through $E - V$                                   | Absent from additive derivative                       |
| Input resistance      | State-dependent, $R^{\text{tot}} = (g^{\text{tot}})^{-1}$ | Effectively fixed                                     |
| Credit interpretation | Inhibition changes path gain and voltage sensitivity      | Inhibition shifts voltage but does not gate path gain |

*Practical heuristic wrappers: 4F and 5F.* **4F (branch-soma relevance proxy).** In the conductance-stage recursion (4), the path gain  $\alpha_n^{\text{cond}}$  attenuates the somatic error differently at each compartment. To compensate without explicitly computing that gain, 4F uses an online branch-soma covariance proxy

$$r_n^{4F} = \frac{\text{Cov}_B(\bar{V}_{n,b}, \bar{V}_{0,b})}{\sqrt{\text{Var}_B(\bar{V}_{n,b}) \text{Var}_B(\bar{V}_{0,b}) + \varepsilon}}.$$

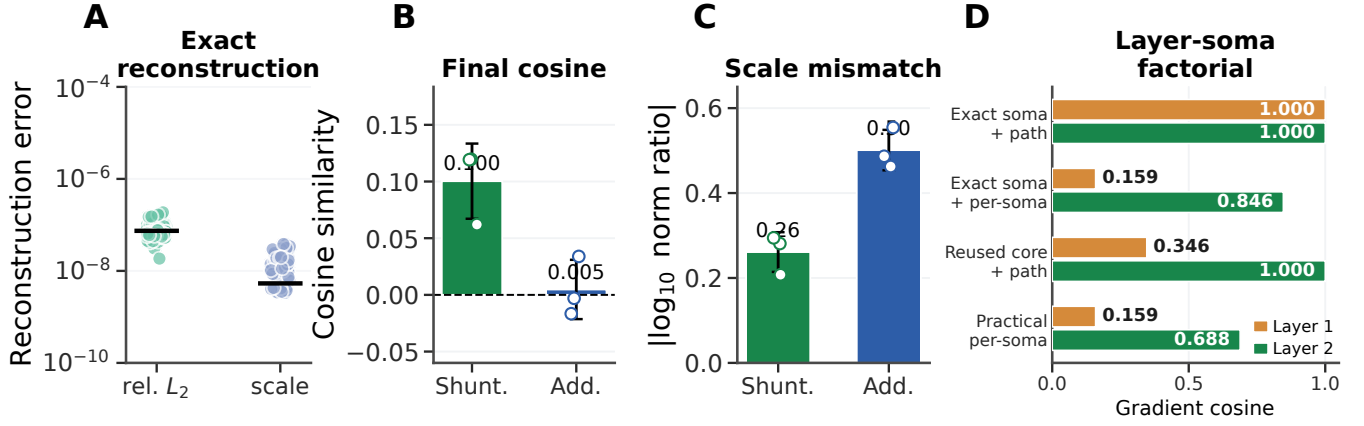


Figure 2: **Exact reconstruction and gradient diagnostics.** (A) Exact-factorization reconstruction matches autograd to numerical precision. (B) Final weighted cosine similarity between LocalCA and backpropagation gradients, reported as a diagnostic rather than the main mechanism claim; dots show seeds. (C) Final scale mismatch,  $|\log_{10}(\|g^{\text{local}}\|/\|g^{\text{bp}}\|)|$ ; dots show seeds. (D) Layer-soma factorial diagnostic at fixed two-layer shunting checkpoints. Bars show energy-weighted 3F conductance/synapse gradient cosines to autograd across six checkpoints; conditions separate exact vs. reused final-core teaching signals and exact path transport vs. per-soma sharing. “Practical per-soma” combines the reused final-core teaching vector with the implemented per-soma feedback construction. Error bars in (B,C) are  $\pm 1$  s.d. across 3 seeds.

Here  $\bar{V}_{n,b}$  is the mean voltage of branch layer  $n$  for batch item  $b$  after averaging over neuron, branch, and spatial coordinates recorded in that layer;  $\bar{V}_{0,b}$  is the corresponding per-example mean soma/core activity. The covariance and variance are estimated over mini-batch examples, then smoothed with an EMA initialized from the first observed value. These statistics are detached from autograd and are updated from the same mini-batch forward activations used for the local gradient. Positive  $r_n^{4F}$  up-weights branch layers whose voltages covary with the soma; weak or sign-opposed covariance reduces the reliability of the shared broadcast for that branch layer. In the implementation used for these experiments the online estimate is clamped to the positive stability range  $[0.1, 2.0]$ , so sign-opposed covariance reduces the multiplier rather than reversing the update direction. Because the numerator and denominator are smoothed separately and may also use single-sample EMA fallbacks, the clamped quantity is a branch-level preconditioner rather than a literal Pearson correlation bounded by one. We keep 4F as a diagnostic intermediate rather than a theorem-derived path-gain estimate. Empirically, this factor alone gives little improvement over 3F; the practical jump comes from the 5F preconditioner below.

**5F (bounded branch-level preconditioner).** 5F adds a second heuristic factor

$$\phi_n = \text{Var}(V_n) / (\sigma_{\text{res}}^2 + \epsilon),$$

where  $\sigma_{\text{res}}^2$  is the online residual variance of the branch-layer voltage after a scalar ridge regression on the recorded parent proxy. In the batch estimator,  $V_n$  and the parent proxy are centered over examples and flattened over the remaining layer coordinates before the smoothed variance and covariance moments are updated. The same detached mini-batch activations are used for statistics and gradients. Operationally,  $\phi_n$  is clamped to  $[0.25, 4.0]$  for stability. This factor is empirical, not theorem-derived; its gain should be read as practical denoising rather than theoretical necessity. With ordinary least squares fit and evaluated on the same observations, the residual variance would not exceed  $\text{Var}(V_n)$ . Here the numerator, parent covariance, residual estimate, and ridge term are updated online and clamped separately, so the ratio can fall below one or rise above one transiently. The lower clamp prevents noisy residual estimates from nearly zeroing a branch-layer update, and the upper clamp prevents a predictable branch from dominating optimization. The ablations in Fig. 5 and the sensitivity check in Appendix Fig. S2 are therefore essential parts of the evidence.

**Definition 2 (5F Update).**

$$\hat{\nabla}_{s_j^{\text{syn}}} L = r_n^{4F} \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n^V \rangle_B, \quad \hat{\nabla}_{s_j^{\text{den}}} L = r_n^{4F} \phi_n \langle R_n^{\text{tot}} (a_j - V_n) e_n^V \rangle_B. \quad (14)$$



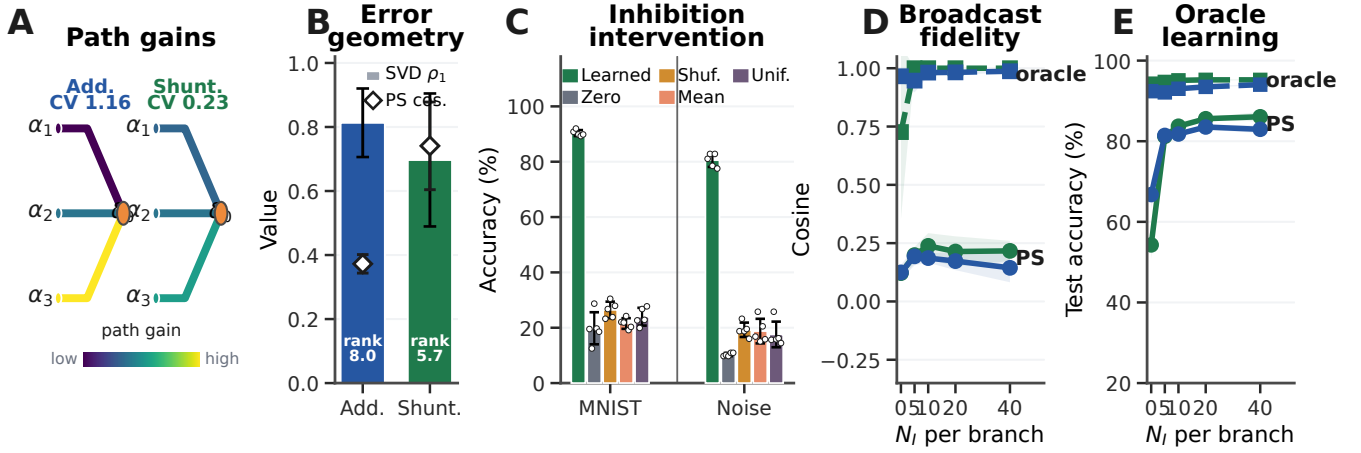
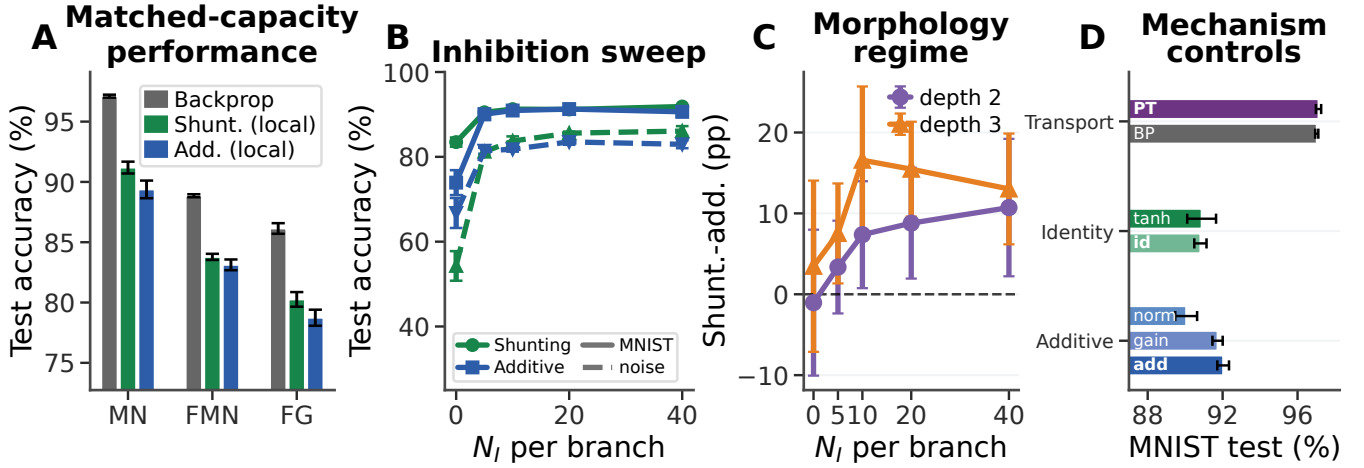


Figure 3: **Mechanistic chain from path gains to learning.** (A) Representative measured conductance-stage path-gain field at matched inhibition. (B) MNIST exact-error geometry: bars show unrestricted rank-1 SVD residual and diamonds show cosine between the actual per-soma broadcast field and the exact compartment-error field. (C) Post-training inhibitory-conductance interventions (learned, zero, shuffled, mean-clamped, or uniform matched inhibition). (D) Compartment-error fidelity on noise resilience; dashed lines show the transported-error oracle. (E) Transported-error learning where per-soma broadcast leaves a gap. Error bars are  $\pm 1$  s.d.

Eq. (14) is therefore the main practical LocalCA update. The 3F rule uses the theorem-derived local eligibility with an approximate broadcast error, while 4F and 5F add empirical reliability factors. A feedback-alignment-style random-broadcast argument is included in Appendix C; it is supportive but not central to the main mechanism claim.

## 4 Experiments

**Setup.** The experiments test five linked predictions: exact factorization should reconstruct autograd gradients; shunting should improve the scale and structure of LocalCA credit signals even when directional cosine is weak; exact compartment errors should have lower intrinsic rank and better alignment with the feedback field used by the learning rule under shunting than under matched additive controls; learned inhibition should be functionally important and sample-dependent rather than merely adding a matched average conductance load; and replacing the broadcast with effective transported error should substantially reduce the local-learning gap. We evaluate matched-capacity supervised tasks and controlled sweeps that isolate inhibition strength, rule family, broadcast mode, morphology, and architecture. The main text uses MNIST [43], Fashion-MNIST [42], figure-ground MNIST, noise resilience, and a compressed morphology-regime summary; CIFAR-10 [44] and routed-feedback diagnostics are reported in the appendix (Figs. S8, S12). Noise resilience is a nonnegative MNIST corruption task with fresh Gaussian latent draws passed through a fixed projection and clipped to  $[0, 1]$ . Figure-ground MNIST places digit signal and distractor pixels in different halves of the input while preserving nonnegative drive; cue-routing in the appendix is the stronger context-dependent branch-credit task. Main comparisons are architecture-matched additive vs. shunting dendritic cores and local vs. backprop training; DFA is a supplementary baseline. Unless otherwise stated, local runs use 5F with per-soma feedback: each soma coordinate supplies the shared teaching signal for that neuron’s compartments. Global scalar, random low-rank, pathway-structured, and transported-error controls are labeled separately. All main-claim dendritic runs use nonnegative conductance transforms and nonnegative first-layer synaptic drive; image inputs are in  $[0, 1]$ , corrupted inputs are clipped back to  $[0, 1]$ , and figure-ground MNIST uses a nonnegative distractor-half construction. For figure-ground MNIST, the main LocalCA competence runs use an HSIC auxiliary term with weight 0.01; the appendix reports the HSIC ablation. The transported-error condition is an oracle upper bound derived from Theorem 1, not the proposed biological rule. Main mechanistic, oracle, and competence results use 5 seeds per condition unless noted otherwise; Appendix A defines the weighted gradient, path-gain CV, and exact-error SVD metrics.



**Figure 4: Matched-capacity performance and regime dependence.** (A) Matched backprop vs. per-soma 5F LocalCA on MNIST, Fashion-MNIST, and figure-ground MNIST. (B) Inhibitory dose-response. (C) Morphology-dependent shunting advantage on noise resilience. (D) MNIST mechanism controls: BP = matched shunting backprop reference, PT = exact path-transported-error LocalCA, tanh/id = learnable bounded tanh vs. identity upward transfer, and add/gain/norm = plain additive, best additive-gain, and best normalized-additive controls. Error bars denote  $\pm 1$  s.d. where available.

*From exact factorization to credit-signal fidelity.* To test whether these performance differences correspond to better credit signals, we compare local and backprop gradients on the same batch at the same parameter values. For each parameter tensor  $p$ , we compute directional alignment (cosine similarity) and scale mismatch, defined as  $|\log_{10}(\|g_p^{\text{local}}\|/\|g_p^{\text{bp}}\|)|$ , aggregated by parameter count.

Across the MNIST gradient-dynamics diagnostic, shunting is closer to backprop than the additive control at the final checkpoint, but the directional agreement is still weak in absolute terms: weighted cosine rises from  $0.005 \pm 0.026$  to  $0.100 \pm 0.033$ , and scale mismatch falls from  $0.50 \pm 0.05$  to  $0.26 \pm 0.05$ . Figure 2A confirms that the implementation matches the theory under the nonnegative-input setup: reconstructing gradients from Corollary 1 using exact compartment errors yields weighted relative reconstruction error below  $2 \times 10^{-7}$  and weighted scale mismatch below  $4 \times 10^{-8}$  across runs. In a few low-norm parameter blocks, cosine becomes numerically unstable even though the reconstruction itself is essentially exact, so we report the norm-based diagnostics in the figure. Figure 2D then isolates the two approximations that enter practical multi-layer LocalCA. With exact layer-soma errors and analytic path transport, 3F local conductance/synapse gradients recover autograd in both layers (energy-weighted cosine 1.000). Compressing the exact layer-soma error into a per-soma shared field mainly damages the first layer (0.159 vs. 0.846), and reusing the final-core teaching vector still leaves the first layer weak even with exact within-tree transport (0.346). The implemented per-soma/reused condition is therefore the practical endpoint of two distinct approximations, not a failure of the exact tree recursion. Appendix Fig. S4 verifies that the corresponding trajectory gradients remain finite, ruling out a trivial zero-gradient explanation for weak cosine diagnostics. At the level of the 3F rule, the central non-local approximation is therefore the broadcast substitute for  $\partial L/\partial V_n$ .

*Mechanistic chain: path gains, broadcast fidelity, and oracle transport.* Figure 3 provides the central empirical chain. At matched inhibition ( $N_I=5$ ), shunting narrows the MNIST conductance-stage path-gain component (CV 0.23 vs. 1.16 for additive; the exact effective gain additionally includes reactivation derivatives) and lowers the intrinsic rank of the exact compartment-error matrix: the unrestricted best rank-1 residual drops from  $0.81 \pm 0.11$  to  $0.70 \pm 0.21$ , while participation rank drops from  $8.0 \pm 3.7$  to  $5.7 \pm 4.0$ . More importantly for the implemented learning rule, the actual per-soma broadcast cosine rises from  $0.37 \pm 0.18$  to  $0.74 \pm 0.13$ . Panel A is an architectural path-gain diagnostic rather than a direct validation of Prop. 3, whose exact expression uses an eligibility-weighted effective-gain dispersion for single-example gradient contributions. The SVD result measures intrinsic geometry; the per-soma broadcast cosine measures compatibility with the fixed feedback field used by the rule. We therefore pair these with the direct broadcast-fidelity and transported-oracle diagnostics in Fig. 3D,E, and we additionally report shared-field

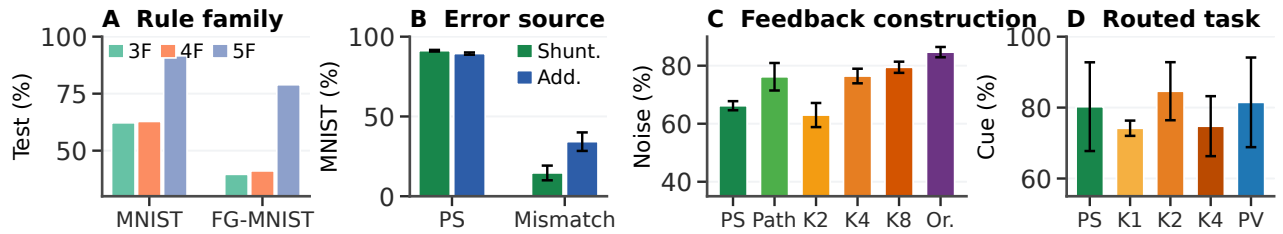


Figure 5: **Rule and feedback controls.** (A) Eligibility-rule ablation: 3F and 4F learn weakly, while 5F is the practical LocalCA rule. (B) Error-source ablation: PS = soma-derived per-soma/shared feedback, and Mismatch = local-mismatch heuristic. (C) Noise-resilience feedback-construction ladder: PS = per-soma/shared baseline, Path = recursive path-propagation proxy, K2/K4/K8 = random low-rank broadcast rank, and Or. = exact transported-error oracle. (D) Cue-routing rank diagnostic: PS = per-soma/shared baseline, K1/K2/K4 = random low-rank ranks, and PV = pathway-vector feedback.

and actual-broadcast residuals for the feedback modes used in the learning rule.

The inhibition intervention shows that the trained computation depends on the learned inhibitory field, rather than only on a matched average conductance load: zeroing, shuffling, or clamping learned branch inhibition drops MNIST from about 90% to 20–26% and noise resilience from about 81% to 10–19%. The fidelity and oracle panels probe the same broadcast bottleneck: per-soma feedback tracks the exact compartment error better under shunting once inhibition is present, transported error reaches about 0.95–1.0 fidelity, and transported learning lifts both the shunting and additive networks to 92–95% on noise resilience. CIFAR-10 and direct-I versus explicit-I path-gain probes remain appendix diagnostics (Fig. S8; Table S5).

*Matched-capacity performance.* In the matched-capacity regime, shunting LocalCA remains within 5–6 percentage points of matched backpropagation on MNIST, Fashion-MNIST, and figure-ground MNIST (Table S3; Fig. 4). The mechanism controls in Fig. 4D sharpen this interpretation: exact path transport reaches the matched shunting backprop reference on MNIST across 3F/5F and decoder-update choices, the identity-transfer version of the shunting core matches the learnable bounded tanh version, and additive gain/normalization variants do not outperform the additive 5F per-soma baseline. These results indicate that feedback-field fidelity is a major bottleneck in this setting, not a generic additive gain or voltage-normalization effect. Fig. 5 summarizes the broader rule and feedback controls: 5F is the strongest practical LocalCA variant, soma-derived feedback is essential, and the noise-resilience rank bridge improves as feedback construction changes from low-rank  $K=1$  to  $K=8$ , with path transport highest.

*Regime dependence of the shunting advantage.* With matched per-soma feedback, shunting is modestly better on MNIST, Fashion-MNIST, and figure-ground MNIST, including  $0.803 \pm 0.006$  vs.  $0.787 \pm 0.007$  for additive on figure-ground MNIST. The gap grows on inhibition-sensitive noise resilience once inhibitory conductance is present (Fig. 4B). Morphology shifts the useful inhibitory operating regime: different tree depths and branch factors peak at different  $N_I$ , and the shunting advantage is not monotonic in inhibition (Fig. 4C). Appendix Fig. S6 extends this boundary check to depth scaling and noisy error broadcasts.

## 5 Discussion

We studied local credit assignment in conductance-based dendritic networks with explicit E/I conductance synapses, shunting inhibition, and tree-structured branching. Exact gradients separate synapse-local eligibility–presynaptic activity, driving force, and input resistance—from one path-specific compartment error, making LocalCA an error-approximation problem. The experiments support a coherent mechanism: branching creates credit paths, E/I synapse banks set the conductance state, and shunting modulates input resistance along those paths. In the regimes studied here this lowers intrinsic exact-error rank, improves direct feedback-fidelity diagnostics, reduces gradient-scale mismatch, and lets transported-error oracles substantially reduce the local-learning gap. Post-training inhibition interventions further show that learned inhibitory conductance is sample-dependent and functionally important to the trained forward computation.

The operating regime matters. Shunting is not universally beneficial, more inhibition is not always better, and the practical 5F rule is an empirical bounded preconditioner. Exact decoder-to-soma feedback is available only when the decoder Jacobian or linear readout supplies the correct soma/core coordinate; practical multi-layer LocalCA reuses neuron coordinates at earlier matched-width layers and therefore also approximates layer-soma errors. The core theorem covers conductance-stage branch parameters under steady-state, non-spiking dynamics; reactivation, decoder, router, top- $k$ , 4F/5F batch-time statistics, and HSIC terms are documented as implementation-specific updates. Eq. (9) is interpretive, and activation shape, morphology, task structure, and feedback rank can all change broadcast compatibility. The inhibition studied here is input-driven rather than recurrent lateral inhibition, and post-training interventions establish forward necessity of the learned inhibitory field rather than proving that those interventions isolate credit-assignment causality. We therefore claim a credit-geometry mechanism supported by exact reconstruction, fidelity diagnostics, controls, and oracle transport, not competitive CIFAR-10 performance, recurrent inhibitory circuits, or sufficiency of a single global scalar broadcast for routed/pathway-specific tasks.

## References

- [1] Koch, C. (1999). *Biophysics of Computation: Information Processing in Single Neurons*. Oxford University Press.
- [2] Abbott, L. F., & Dayan, P. (2001). *Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems*. MIT Press.
- [3] Poirazi, P., Brannon, T., & Mel, B. W. (2003). Pyramidal neuron as two-layer neural network. *Neuron*, 37(6), 989–999.
- [4] London, M., & Häusser, M. (2005). Dendritic computation. *Annual Review of Neuroscience*, 28, 503–532.
- [5] Spruston, N. (2008). Pyramidal neurons: dendritic structure and synaptic integration. *Nature Reviews Neuroscience*, 9, 206–221. <https://doi.org/10.1038/nrn2286>
- [6] Branco, T., & Häusser, M. (2010). The single dendritic branch as a fundamental functional unit in the nervous system. *Current Opinion in Neurobiology*, 20(4), 494–502. <https://doi.org/10.1016/j.conb.2010.07.009>
- [7] Larkum, M. (2013). A cellular mechanism for cortical associations: an organizing principle for the cerebral cortex. *Trends in Neurosciences*, 36(3), 141–151. <https://doi.org/10.1016/j.tins.2012.11.006>
- [8] Urbanczik, R., & Senn, W. (2014). Learning by the dendritic prediction of somatic spiking. *Neuron*, 81(3), 521–528.
- [9] Carandini, M., & Heeger, D. J. (2012). Normalization as a canonical neural computation. *Nature Reviews Neuroscience*, 13(1), 51–62.
- [10] Holt, G. R., & Koch, C. (1997). Shunting inhibition does not have a divisive effect on firing rates. *Neural Computation*, 9(5), 1001–1013.
- [11] Vogels, T. P., Sprekeler, H., Zenke, F., Clopath, C., & Gerstner, W. (2011). Inhibitory plasticity balances excitation and inhibition in sensory pathways and memory networks. *Science*, 334(6062), 1569–1573.
- [12] Frémaux, N., & Gerstner, W. (2016). Neuromodulated spike-timing-dependent plasticity, and theory of three-factor learning rules. *Frontiers in Neural Circuits*, 9, 85. <https://doi.org/10.3389/fncir.2015.00085>
- [13] Hennequin, G., Agnes, E. J., & Vogels, T. P. (2017). Inhibitory plasticity: balance, control, and codependence. *Annual Review of Neuroscience*, 40, 557–579. <https://doi.org/10.1146/annurev-neuro-072116-031005>
- [14] Lillicrap, T. P., Cownden, D., Tweed, D. B., & Akerman, C. J. (2016). Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications*, 7, 13276.

- [15] Lillicrap, T. P., Santoro, A., Marris, L., Akerman, C. J., & Hinton, G. (2020). Backpropagation and the brain. *Nature Reviews Neuroscience*, 21(6), 335–346. <https://doi.org/10.1038/s41583-020-0277-3>
- [16] Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, 323, 533–536.
- [17] Nøkland, A. (2016). Direct feedback alignment provides learning in deep neural networks. *NeurIPS*, 29, 1037–1045.
- [18] Guerguiev, J., Lillicrap, T. P., & Richards, B. A. (2017). Towards deep learning with segregated dendrites. *eLife*, 6, e22901.
- [19] Sacramento, J., Costa, R. P., Bengio, Y., & Senn, W. (2018). Dendritic cortical microcircuits approximate the backpropagation algorithm. *NeurIPS*, 31.
- [20] Whittington, J. C., & Bogacz, R. (2019). Theories of error back-propagation in the brain. *Trends in Cognitive Sciences*, 23(3), 235–250.
- [21] Richards, B. A., & Lillicrap, T. P. (2019). Dendritic solutions to the credit assignment problem. *Current Opinion in Neurobiology*, 54, 28–36.
- [22] Scellier, B., & Bengio, Y. (2017). Equilibrium Propagation: Bridging the Gap between Energy-Based Models and Backpropagation. *Frontiers in Computational Neuroscience*, 11, 24. <https://doi.org/10.3389/fncom.2017.00024>
- [23] Song, Y., Millidge, B., Salvatori, T., Lukasiewicz, T., Xu, Z., & Bogacz, R. (2024). Inferring neural activity before plasticity as a foundation for learning beyond backpropagation. *Nature Neuroscience*, 27, 348–358. <https://doi.org/10.1038/s41593-023-01514-1>
- [24] Gretton, A., Bousquet, O., Smola, A. J., & Schölkopf, B. (2005). Measuring statistical dependence with Hilbert-Schmidt norms. *Lecture Notes in Computer Science*, 3734, 63–77. [https://doi.org/10.1007/11564089\\_7](https://doi.org/10.1007/11564089_7)
- [25] Welford, B. P. (1962). Note on a method for calculating corrected sums of squares and products. *Technometrics*, 4(3), 419–420. <https://doi.org/10.1080/00401706.1962.10490022>
- [26] Turrigiano, G. G. (2008). The self-tuning neuron: synaptic scaling of excitatory synapses. *Cell*, 135(3), 422–435.
- [27] Payeur, A., Guerguiev, J., Zenke, F., Richards, B. A., & Naud, R. (2021). Burst-dependent synaptic plasticity can coordinate learning in hierarchical circuits. *Nature Neuroscience*, 24(7), 1010–1019.
- [28] Greedy, W., Zhu, H. W., Pemberton, J., Mellor, J., & Ponte Costa, R. (2022). Single-phase deep learning in cortico-cortical networks. *NeurIPS*, 35, 24213–24225.
- [29] Haider, P., Ellenberger, B., Kriener, L., Jordan, J., Senn, W., & Petrovici, M. A. (2021). Latent Equilibrium: A unified learning theory for arbitrarily fast computation with arbitrarily slow neurons. *NeurIPS*, 34, 17839–17851.
- [30] Hinton, G. (2022). The Forward-Forward Algorithm: Some Preliminary Investigations. *arXiv:2212.13345*.
- [31] Dellaferrera, G., & Kreiman, G. (2022). Error-driven input modulation: Solving the credit assignment problem without a backward pass. *Proceedings of the 39th International Conference on Machine Learning, Proceedings of Machine Learning Research*, 162, 4937–4955.
- [32] Lee, D.-H., Zhang, S., Fischer, A., & Bengio, Y. (2015). Difference target propagation. *Machine Learning and Knowledge Discovery in Databases: ECML PKDD 2015*, 498–515. [https://doi.org/10.1007/978-3-319-23528-8\\_31](https://doi.org/10.1007/978-3-319-23528-8_31)
- [33] Meulemans, A., Tristany Farinha, M., García Ordóñez, J., Vilimelis Aceituno, P., Sacramento, J., & Grewe, B. F. (2021). Credit assignment in neural networks through deep feedback control. *NeurIPS*, 34, 4674–4687.

- [34] Millidge, B., Seth, A. K., & Buckley, C. L. (2021). Predictive coding: A theoretical and experimental review. *arXiv:2107.12979*.
- [35] Koch, C., Poggio, T., & Torre, V. (1983). Nonlinear interactions in a dendritic tree: localization, timing, and role in information processing. *PNAS*, 80(9), 2799–2802. <https://doi.org/10.1073/pnas.80.9.2799>
- [36] Silver, R. A. (2010). Neuronal arithmetic. *Nature Reviews Neuroscience*, 11(7), 474–489.
- [37] Max, K., Kriener, L., Pineda García, G., Nowotny, T., Jaras, I., Senn, W., & Petrovici, M. A. (2024). Learning efficient backprojections across cortical hierarchies in real time. *Nature Machine Intelligence*, 6, 619–630. <https://doi.org/10.1038/s42256-024-00845-3>
- [38] Ma, C., Wu, J., Si, C., & Tan, K. C. (2024). Scaling supervised local learning with augmented auxiliary networks. *International Conference on Learning Representations (ICLR)*.
- [39] Lv, C., Xu, J., Lu, Y., Wang, X., Wang, Z., Xu, Z., Yu, D., Du, X., Zheng, X., & Huang, X. (2025). Dendritic Localized Learning: Toward Biologically Plausible Algorithm. *Proceedings of the 42nd International Conference on Machine Learning, Proceedings of Machine Learning Research*, 267, 41682–41700.
- [40] Erdogan, M., Pehlevan, C., & Erdogan, A. T. (2025). Error Broadcast and Decorrelation as a Potential Artificial and Natural Learning Mechanism. *Advances in Neural Information Processing Systems*, 38.
- [41] Kao, C.-H., & Hariharan, B. (2024). Counter-Current Learning: A Biologically Plausible Dual Network Approach for Deep Learning. *Advances in Neural Information Processing Systems*, 37.
- [42] Xiao, H., Rasul, K., & Vollgraf, R. (2017). Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms. *arXiv:1708.07747*.
- [43] LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278–2324.
- [44] Krizhevsky, A. (2009). Learning multiple layers of features from tiny images. Technical report, University of Toronto.
- [45] Francioni, V., Tang, V. D., Toloza, E. H. S., Ding, Z., Brown, N. J., & Harnett, M. T. (2026). Vectorized instructive signals in cortical dendrites. *Nature*, 652, 1254–1263. <https://doi.org/10.1038/s41586-026-10190-7>
- [46] Iyer, A., Grewal, K., Velu, A., Souza, L. O., Forest, J., & Ahmad, S. (2022). Avoiding catastrophe: Active dendrites enable multi-task learning in dynamic environments. *Frontiers in Neurobotics*, 16, 846219.

## A Supplementary Results and Diagnostics

This appendix records the supporting diagnostics, implementation details, and task protocols needed to interpret and reproduce the paper’s main claims.

**Appendix structure.** The appendix is organized to make the paper reproducible from mechanism to implementation. Appendix A first reports the supplementary diagnostics that support or delimit the main empirical claims. Appendix B then specifies the implemented dendritic networks, including morphology, synapse banks, shunting and additive forward equations, positive parameterization, reactivation, decoder errors, and optimizer-level gradient assignment. Appendix C collects learning-rule variants and auxiliary theory details that are useful for reproducing the local updates but are not needed for the main proof. Appendix D gives task construction and robustness protocols.

**Boundary diagnostics.** Several controls are intentionally negative or mixed. The log-gain covariance margin from Eq. (9) is not used as a positive main result, and the pathway-vector feedback variant does not beat the best unstructured low-rank cue-routing control. These diagnostics delimit the claim rather than smoothing over failure modes.

## Reproducibility Summary

Detailed run sources and commands are maintained in REPRODUCIBILITY.md. The table below summarizes the experiments, architectures, seed protocols, selection rules, and reproduction commands used for the manuscript figures.

| Experiment family                                           | Architecture / condition                                                                            | Seeds and selection                                                              | Reproduction command                                               |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Exact-factorization and gradient diagnostics                | Nonnegative-input E/I dendritic cores, [128, 128] soma layers, [3, 3] tree, 3F–5F local rules       | Main summaries use 5 seeds; gradient-dynamics panels use 3 seeds where indicated | <code>python scripts/generate_neurips_figures.py</code>            |
| Layer-soma factorial diagnostic                             | Two-layer shunting checkpoints with exact/reused soma errors and transported/per-soma branch fields | Six fixed checkpoints; no validation selection                                   | <code>python scripts/measure_layer_soma_factorial.py</code>        |
| Path-gain, exact-error rank, and inhibition intervention    | Matched additive and shunting cores at fixed inhibition; post-training intervention states          | Five seeds per main condition; interventions use trained checkpoints             | <code>python scripts/generate_theory_diagnostics_figures.py</code> |
| Transported-error oracle and feedback construction controls | Per-soma, random low-rank, path-structured, and exact transported-error feedback                    | Five seeds for main oracle and feedback-ladder summaries                         | <code>python scripts/summarize_path_transport_sweep.py</code>      |
| Matched-capacity performance                                | MNIST, Fashion-MNIST, figure-ground MNIST; per-soma 5F LocalCA and matched backprop references      | Five seeds; validation-selected configurations report held-out test accuracy     | <code>python scripts/summarize_neurips_claim_sweeps.py</code>      |

Table S1: **Reproducibility summary.** Commands are run from the draft directory after the corresponding training or diagnostic outputs have been generated.

## Diagnostic Metrics

The gradient-fidelity diagnostics compare LocalCA and backpropagation gradients at matched parameters and batches. For a parameter block  $p$ , let  $g_p^{\text{local}}$  and  $g_p^{\text{bp}}$  be the two gradients and let  $w_p = \text{numel}(p) / \sum_q \text{numel}(q)$ . The reported weighted cosine is

$$C_w = \sum_p w_p \frac{\langle g_p^{\text{local}}, g_p^{\text{bp}} \rangle}{\|g_p^{\text{local}}\|_2 \|g_p^{\text{bp}}\|_2 + \epsilon},$$

and weighted scale mismatch is

$$S_w = \sum_p w_p \left| \log_{10} \frac{\|g_p^{\text{local}}\|_2 + \epsilon}{\|g_p^{\text{bp}}\|_2 + \epsilon} \right|.$$

For exact-error compressibility, we form a matrix  $E$  whose rows are examples and whose columns are compartment–neuron coordinates of the exact pre-reactivation compartment error. If  $E = U\Sigma V^\top$  and  $\sigma_i$  are the singular values, the unrestricted rank-1 residual and participation rank are

$$\rho_1 = \frac{\sqrt{\sum_{i>1} \sigma_i^2}}{\sqrt{\sum_i \sigma_i^2}}, \quad r_{\text{part}} = \frac{(\sum_i \sigma_i^2)^2}{\sum_i \sigma_i^4}.$$

This is an intrinsic geometry metric: both the sample coefficients and spatial template are chosen in hindsight by SVD. The constrained shared-field residual instead fixes the spatial template to all ones and allows only the best per-example scalar,

$$\rho_1 = \frac{\|E - \text{rowmean}(E)\mathbf{1}^\top\|_F}{\|E\|_F}.$$

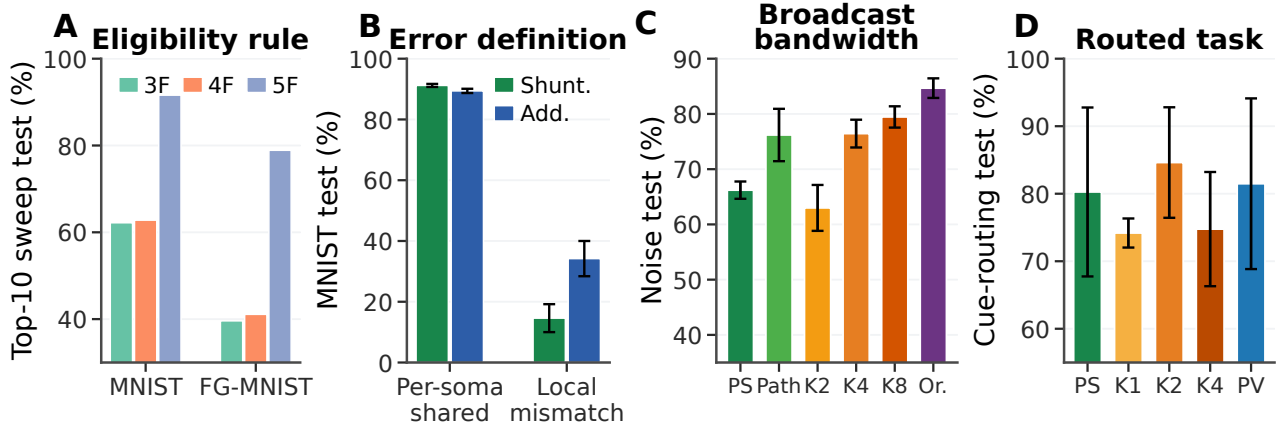


Figure S1: **Detailed rule and feedback controls.** (A) Eligibility-rule ablation: 3F, 4F, and 5F. (B) Error-source ablation: per-soma/shared feedback vs. local mismatch. (C) Feedback-bandwidth ladder on noise resilience: PS per-soma/shared, recursive Path, K2/K4/K8, and transported Oracle. (D) Cue routing: PS per-soma/shared, random low-rank K1/K2/K4, and pathway-vector feedback.

The per-soma/blockwise residual matches the practical feedback object more closely. Let  $\Pi_{PS}$  replace, for each sample and soma, all compartments of that soma by their optimal shared coefficient. Then

$$\rho_{PS} = \frac{\|E - \Pi_{PS}E\|_F}{\|E\|_F}.$$

The broadcast-conditioned residual uses the actual feedback field  $B$  generated by the implemented mode,

$$\rho_{bc} = \frac{\|E - B\|_F}{\|E\|_F},$$

with a corresponding flattened cosine and, where useful, a globally rescaled version of  $B$ . These constrained metrics are emitted by `scripts/measure_error_rank_diagnostics.py` alongside the unrestricted SVD metrics, so intrinsic error geometry and compatibility with the implemented broadcast field can be reported separately. Path-gain dispersion is the coefficient of variation of  $\alpha_n^{\text{cond}}$  or  $\tilde{\alpha}_n$  over the measured sample and compartment field, depending on whether reactivation is enabled. This unweighted architectural statistic differs from the eligibility-weighted  $CV_w(\tilde{\alpha}_{n(i)})$  in Prop. 3. Compartment-error fidelity is the cosine between the tested broadcast field and the exact compartment-error field after flattening over the same sample and compartment axes.

## Rule and Feedback Controls

These controls separate three questions that are deliberately collapsed in the main competence summary: which local-rule wrapper is practical, whether the somatic error source matters, and how performance changes as the feedback field gains bandwidth or path information.

## Rule-Family Ranking

This table keeps the 3F/4F/5F optimizer distinction explicit: the theorem-derived eligibility is central to the proposed mechanism, but the reported LocalCA performance results use the empirical 5F stabilizer.

## 5F Stabilizer Sensitivity

Because the main LocalCA runs use the empirical 5F stabilizer, we also tested whether performance depends sharply on the preconditioner clamp or the EMA rate used to estimate branch statistics. Across three-seed MNIST sensitivity



| Dataset             | Rule | Mean valid, 10 validation-selected configs | Mean test, same 10 configs |
|---------------------|------|--------------------------------------------|----------------------------|
| MNIST               | 3F   | 0.611                                      | 0.622                      |
| MNIST               | 4F   | 0.620                                      | 0.628                      |
| MNIST               | 5F   | <b>0.912</b>                               | <b>0.916</b>               |
| Figure-ground MNIST | 3F   | 0.398                                      | 0.396                      |
| Figure-ground MNIST | 4F   | 0.411                                      | 0.411                      |
| Figure-ground MNIST | 5F   | <b>0.807</b>                               | <b>0.789</b>               |

Table S2: **Rule-family ranking.** Mean over the ten best validation-selected configurations from local-competence sweeps; this is not top-10 classification accuracy, and test scores are reported only after validation selection.

checks, tightening or widening the clamp and varying the EMA rate leaves test accuracy near the default setting. This does not make 5F theorem-derived; it supports interpreting it as a bounded branch-level reliability preconditioner rather than a brittle tuned error source.

## Local Performance and Regime Dependence

Table S3 reports the matched-capacity performance numbers used in the main text. The point of this table is not to claim state-of-the-art benchmark performance, but to show the remaining LocalCA-to-backprop gap under matched dendritic capacity and per-soma 5F feedback.

## Extended Gradient Analysis and $N_I$ Sweep Detail

Figure 2 in the main text summarizes final-state gradient diagnostics and the layer-soma factorial check. The supplementary view below separates static scale checks, inhibitory dose-response curves, and individual Fashion-MNIST seed behavior so that the aggregate claims are not driven by a single summary panel.

**Alignment dynamics are not a zero-gradient artifact.** To check whether low additive alignment could be caused by vanishing gradients, we analyzed a separate 3-seed checkpoint trajectory with parameter-level LocalCA and backprop gradients. The additive local gradients are nonzero throughout training (weighted norm range  $9.5 \times 10^{-4}$ – $9.5 \times 10^{-3}$ ), and the corresponding backprop gradients are also nonzero ( $2.3 \times 10^{-5}$ – $5.7 \times 10^{-3}$ ). The low additive cosine therefore reflects directional and scale mismatch, not absent gradients: additive 5F cosine stays near zero from epoch 0 to epoch 50 (0.018 to 0.005), while its local/backprop norm ratio falls from about  $8.8 \times 10^2$  to 0.69.

## Verification and Reproducibility

These checks compare the main seed set with held-out seeds and report the HSIC-weight sensitivity for figure-ground MNIST. The additional matched experiments use the same objective-weight comparison on both seed sets, so seed robustness and HSIC contribution can be summarized as a matched two-factor control.

## Additional Stress Tests

These stress tests probe two ways in which low-bandwidth broadcast can become unreliable: deeper dendritic paths and noisy teaching signals. They are not separate main benchmarks; they delimit where shunting remains useful and where additive local learning degrades.

## FA/DFA Baseline Comparison

The feedback-alignment baseline is included as an adjacent local-learning reference point. DFA can be evaluated on the dendritic cores, whereas the FA condition is marked unavailable for the block-structured dendritic updates because the required random feedback matrices are not dimensionally compatible with the branch-level parameterization.

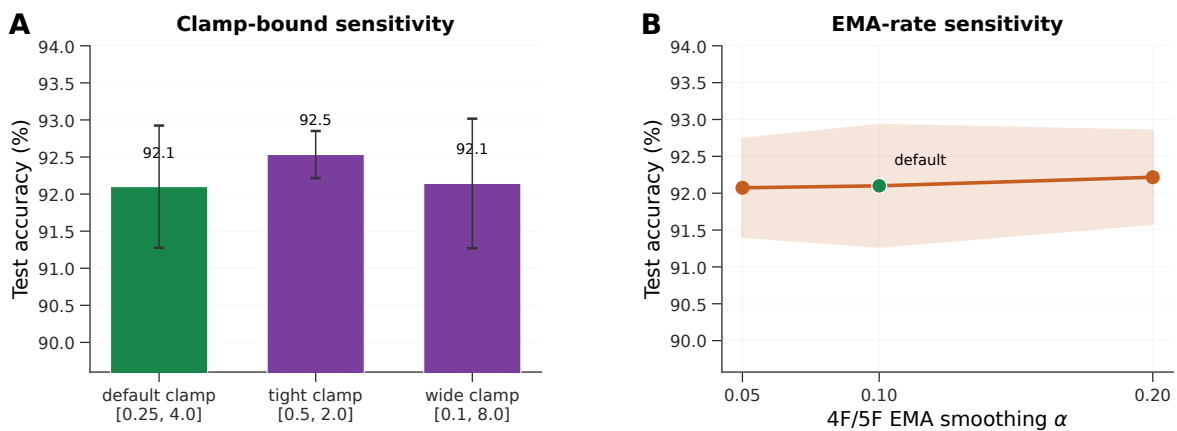


Figure S2: **5F stabilizer sensitivity.** (A) Changing the 5F preconditioner clamp from the default [0.25, 4.0] to tighter [0.5, 2.0] or wider [0.1, 8.0] bounds leaves MNIST test accuracy near 92%. (B) Changing the 4F/5F EMA smoothing rate over  $\alpha \in \{0.05, 0.10, 0.20\}$  gives similarly stable performance. Error bars and bands are  $\pm 1$  s.d. across 3 seeds.

## CIFAR-10 Results

Table S4 provides the exact values for the appendix CIFAR-10 figure. CIFAR-10 is used only as a harder-dataset mechanistic stress test: under flattened inputs, nonnegative conductance constraints, and a compact passive dendritic core, the matched backpropagation reference is the relevant comparison rather than modern vision accuracy. In this stronger compact direct-I-stream family, shunting performs well under standard training (49.5% vs. 48.3% for additive), so the harder-dataset stress test does not unfairly penalize the shunting architecture. The broadcast-fidelity ladder also survives once decoder-aware soma mapping is used: additive LocalCA reaches 25.5% under per-soma feedback and 32.4% under transported error, while shunting rises from 29.7% under per-soma feedback to 35.0% with random low-rank  $K=4$  and 47.2% with transported error, nearly matching the 49.5% shunting backpropagation reference. Removing learned I-to-E conductance lowers the shunting reference by 5.7 pp and transported LocalCA by 8.6 pp, making this a harder-dataset stress test for inhibitory conductance as part of the useful credit geometry rather than just an added architectural detail.

## Additive Normalization Control

This control separates generic additive voltage normalization from shunting conductance dynamics in a lower-capacity MNIST stress regime.

## Morphology $\times$ Inhibition Regime Map

This regime map tests whether tree geometry changes the inhibitory operating point at which shunting helps. It supports the claim that morphology changes the useful inhibition regime; a fuller mechanistic account would require matched theory diagnostics for every cell in the grid.

## Input-Mode Probe: Direct Inhibitory Stream vs. Explicit Inhibitory Cells

The main experiments use input-driven inhibitory streams: the transfer layer provides nonnegative excitatory and inhibitory channels from the external input, and excitatory dendrites receive learned I-to-E conductance banks directly. To test whether the same mechanism can be generated by an explicit feedforward inhibitory population, we ran a one-feedforward-layer noise-resilience probe that preserves the relevant dendritic depth ([3, 3] morphology). In this setting, explicit inhibitory cells reach  $94.2 \pm 0.2\%$  under path-transport LocalCA when their dendrites use the same update policy as excitatory dendrites,  $93.8 \pm 0.2\%$  when those inhibitory-cell updates are frozen, and  $95.1 \pm 0.3\%$

| Dataset             | Shunt. BP | Add. BP           | Shunt. local      | Add. local        | Shunt. gap (pp) |
|---------------------|-----------|-------------------|-------------------|-------------------|-----------------|
| MNIST               | 0.971     | $0.970 \pm 0.001$ | $0.911 \pm 0.005$ | $0.894 \pm 0.007$ | 6.0             |
| Fashion-MNIST       | 0.889     | $0.876 \pm 0.001$ | $0.838 \pm 0.003$ | $0.831 \pm 0.004$ | 5.1             |
| Figure-ground MNIST | 0.861     | –                 | $0.803 \pm 0.006$ | $0.787 \pm 0.007$ | 5.8             |

Table S3: **Local performance.** Backpropagation references are matched-capacity standard-training runs. Additive backpropagation is reported where a matched five-seed reference is available; the figure-ground additive backpropagation reference was not part of the matched competence source table. Local values are 5F with per-soma feedback and local decoder updates. Figure-ground MNIST additionally uses an HSIC auxiliary objective (weight 0.01; see the HSIC auxiliary-objective appendix paragraph). Error bars are  $\pm 1$  s.d. across 5 seeds where available; gaps are percentage points.

under standard backprop. Direct input-driven inhibition gives the expected broadcast-fidelity ladder on the same one-layer architecture: per-soma feedback reaches  $85.2 \pm 0.7\%$ , while exact effective path transport reaches  $95.2 \pm 0.0\%$ . Thus explicit inhibitory cells can generate useful path-gain structure once the architecture isolates the dendritic-path question.

**Post-training inhibition intervention and exact-error rank.** We next tested whether learned inhibitory conductance carries sample-dependent structure rather than only changing global scale. We replayed trained shunting LocalCA checkpoints while replacing the branch-level inhibitory conductance with four post-training interventions: zero inhibition, sample-shuffled inhibition, per-branch batch means, or one uniform matched mean. On a 512-example held-out subset, learned MNIST inhibition at  $N_I=5$  gives  $90.4 \pm 1.1\%$  accuracy, while the four interventions drop to 19.8–26.4%. On noise resilience at  $N_I=10$ , learned inhibition gives  $80.5 \pm 2.4\%$ , while the interventions drop to 10.3–19.3%. Thus the inhibitory field is not interchangeable with a matched global conductance load. We also computed the SVD of exact compartment-error matrices across samples and compartments. At matched MNIST  $N_I=5$ , shunting has lower unrestricted rank-1 residual than additive ( $0.70 \pm 0.21$  vs.  $0.81 \pm 0.11$ ), lower participation rank ( $5.7 \pm 4.0$  vs.  $8.0 \pm 3.7$ ), and higher actual per-soma broadcast cosine ( $0.74 \pm 0.13$  vs.  $0.37 \pm 0.18$ ), supporting but not by itself proving the path-gain compressibility interpretation.

We also computed the direct log-gain covariance margin in Eq. (9) on the selected morphology diagnostic checkpoints. The margin is zero when no inhibitory conductance is present and slightly negative in the inhibited shunting selections ( $-2.5 \times 10^{-3}$ ,  $-5.0 \times 10^{-4}$ , and  $-2.8 \times 10^{-4}$  at  $N_I=5, 20, 40$ ). We therefore do not use this covariance condition as a positive main result. Its role is to clarify when shunting should compress gains; the empirical claim rests on the measured comparative path-gain dispersion, exact-error rank, and inhibition-intervention diagnostics.

## Experimental Predictions

The model is not a detailed cell-type circuit model, but its path-gain identity gives falsifiable dendritic predictions. These predictions are naturally aligned with branch-level dendritic computation and compartment-targeted inhibition in pyramidal neurons [5, 6, 7, 13].

## Cue-Routing Feedback-Rank Diagnostic

Cue routing tests a regime where one shared teaching signal is expected to fail: context determines which noisy cue is reliable, so different pathways should receive different credit. Figure S12 shows that the benchmark is learnable and that moving beyond the restricted per-soma / single-channel baselines helps, but the path-structured variant does not yet beat the best unstructured low-rank setting. We therefore interpret this result as a failure-mode and rank-requirement diagnostic, not as a solved structured-feedback method.

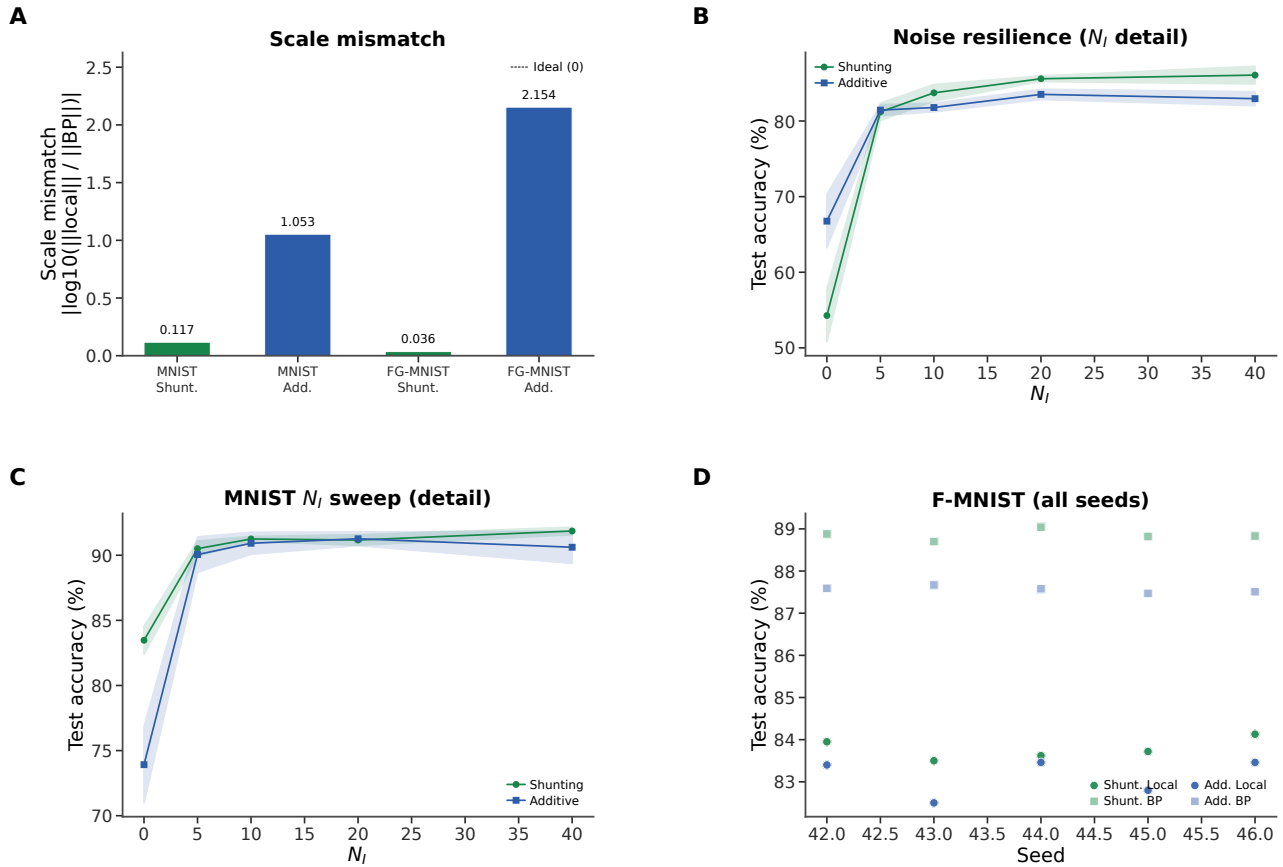


Figure S3: **Extended gradient and  $N_I$  analysis (supplementary).** (A) Log-scale mismatch,  $|\log_{10}(\|g^{\text{local}}\|/\|g^{\text{bp}}\|)|$ , from a separate matched-weight static diagnostic; these values should not be compared directly to the checkpointed final-state diagnostic in Fig. 2. The ideal value is 0; shunting is near ideal (0.117), whereas additive shows order-of-magnitude distortion ( $> 1.0$ ). (B) Noise-resilience  $N_I$  dose-response with error bands ( $\pm 1$  s.d.). (C) MNIST  $N_I$  dose-response detail with error bands. (D) Fashion-MNIST individual seeds for all conditions, showing consistency across runs.

## Random Low-Rank Broadcast Channels

These controls test whether failures of the restricted per-soma baseline reflect insufficient feedback bandwidth rather than a failure of local eligibility itself.

## B Dendritic Network Implementation

This section gives the implementation-level specification needed to reproduce the dendritic architectures and LocalCA updates. It separates the biological modeling assumptions from software choices such as positive raw-parameter transforms, decoder-error coordinates, and optimizer gradient assignment.

### Implementation Overview

Each feedforward dendritic layer is a population of dendritic neurons. A neuron contains a rooted tree of compartments; each compartment has an excitatory synapse bank, optionally an inhibitory synapse bank, and dendritic coupling conductances from its children. The main experiments use branch-level input-driven inhibition: a nonnegative transfer stage produces excitatory and inhibitory input streams from the external input, and the inhibitory stream drives learned I-to-E conductance banks on excitatory dendritic branches. The explicit-I probe in Table S5 instead instantiates a

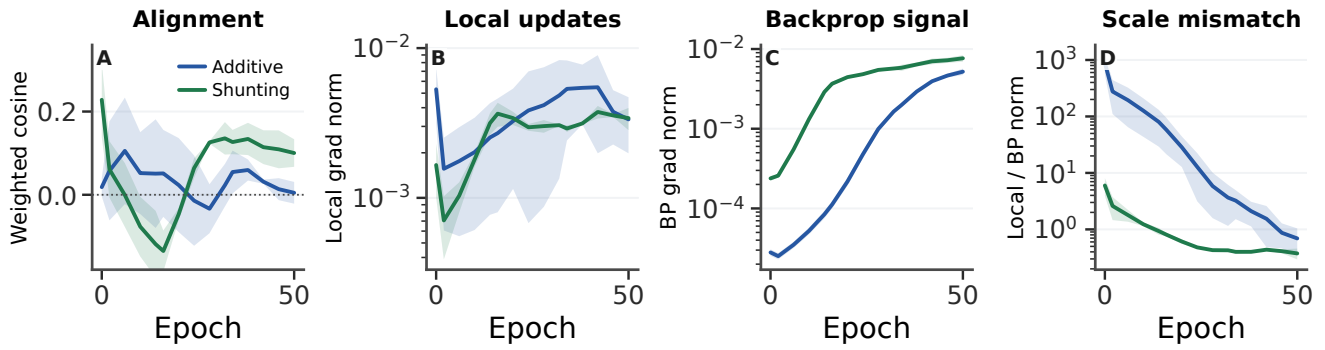


Figure S4: **Alignment dynamics with explicit gradient-norm checks.** (A) Weighted cosine similarity between LocalCA and backpropagation gradients over training for the 5F rule. (B,C) Weighted local and backprop gradient norms stay nonzero for both additive and shunting models. (D) The additive control begins with a much larger scale mismatch, then approaches the backprop scale without recovering strong directional alignment. Shading is s.d. across 3 seeds.

separate feedforward inhibitory population, whose output projects inhibitory conductance onto excitatory dendrites.

The additive and shunting cores are architecture-matched: they use the same tree, synapse counts, nonnegative transfer stage, positive conductance parameterization, optimizer protocol, and seed sets. They differ only in how a branch combines excitatory, inhibitory, and dendritic inputs. Decoder weights are unconstrained readout parameters and are not interpreted as conductances.

| Paper object              | Code anchor                                                                                                                                                                                                                                                                                     | Role                                                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Shunting branch voltage   | <code>DendriticBranchLayer.forward</code> with <code>use_shunting=True</code>                                                                                                                                                                                                                   | Implements numerator/denominator conductance integration and stores total conductance for diagnostics.           |
| Additive control voltage  | <code>DendriticBranchLayer.forward</code> with <code>use_shunting=False</code> ; <code>normalize_additive_voltage</code> for the normalization control                                                                                                                                          | Uses the same branch tree and synapse banks, but inhibitory input is subtractive rather than a denominator load. |
| Broadcast-mode switch     | <code>LocalCreditAssignment._apply_local_rule_gradients</code> ;                                                                                                                                                                                                                                | Chooses scalar/per-soma, local-mismatch, low-rank, path-transport, and pathway-vector feedback fields.           |
| Low-rank feedback         | <code>LocalRuleConfig.error_broadcast_mode</code>                                                                                                                                                                                                                                               | Fixed random low-rank broadcast controls used in the bandwidth ladder.                                           |
| Path-transport oracle     | <code>LocalCreditAssignment._compute_low_rank_broadcast</code> ; <code>broadcast_rank</code>                                                                                                                                                                                                    | Transported-error upper bound using the effective tree recursion.                                                |
| Path-propagation proxy    | <code>LocalCreditAssignment._precompute_path_transport_errors</code>                                                                                                                                                                                                                            | Approximate morphology-aware attenuation control, distinct from exact path transport.                            |
| Shunting eligibility      | <code>LocalCreditAssignment._precompute_path_propagation_factors</code>                                                                                                                                                                                                                         | Assigns $xR^{\text{tot}}(E - V)e$ gradients, then multiplies by the positive-transform derivative.               |
| Additive eligibility      | <code>LocalCreditAssignment._apply_local_rule_gradients</code> conductance branch                                                                                                                                                                                                               | Assigns the matched signed additive derivative, with optional additive-gain controls.                            |
| 4F / 5F reliability gates | <code>LocalCreditAssignment._apply_local_rule_gradients</code> additive branch                                                                                                                                                                                                                  | Slow reliability preconditioners used by 4F/5F, not additional fast task-error channels.                         |
| Experiment knobs          | <code>_compute_layer_rho</code> , <code>_compute_layer_phi*</code> , <code>clamp_phi</code> , <code>FiveFactorConfig</code> , <code>LocalRuleConfig</code> ; YAML fields <code>rule_variant</code> , <code>error_broadcast_mode</code> , <code>broadcast_rank</code> , <code>five_factor</code> | Reproduces the rule family, feedback bandwidth, and 5F stabilizer settings.                                      |

Table S8: **Equation-to-code map.** Code anchors are symbolic names rather than line numbers so that minor refactors do not invalidate the reproducibility map.

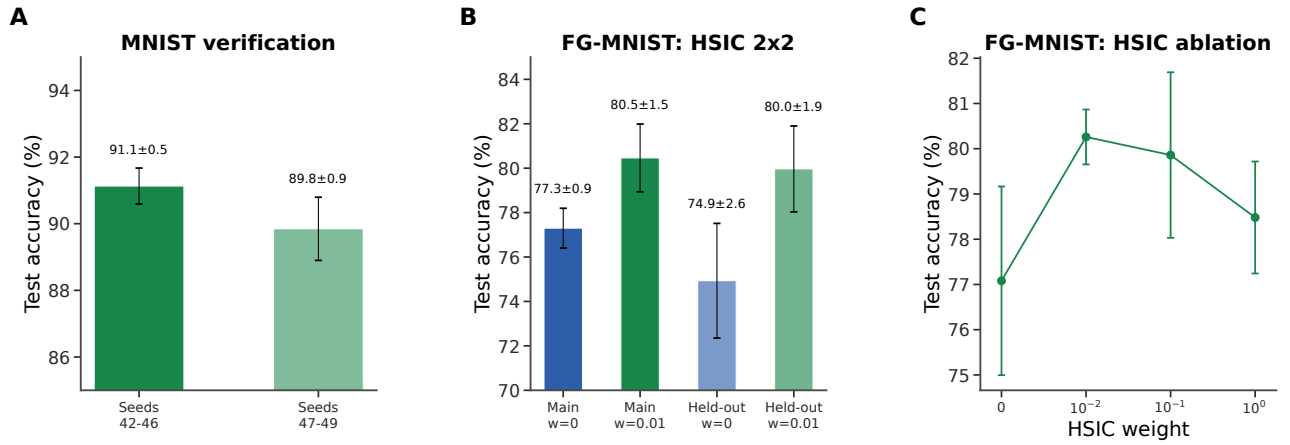


Figure S5: **Verification and reproducibility (supplementary).** (A) MNIST verification: the current-code rerun of main seeds (42–46) yields  $91.1 \pm 0.5\%$ ; held-out seeds (47–49) yield  $89.8 \pm 0.9\%$ , showing robustness across random seeds. (B) Matched figure-ground MNIST HSIC control: both main seeds (42–46) and held-out seeds (47–49) compare HSIC weight 0 against 0.01. (C) HSIC weight ablation on figure-ground MNIST: moderate weights (0.01–0.1) perform best.

| Parameter family                                | Forward role                                | Update source                                                                 | Feedback / auxiliary signal                                                                                            |
|-------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Excitatory and inhibitory synaptic conductances | Branch E/I drive                            | LocalCA conductance eligibility                                               | Scalar, per-soma, low-rank, path-structured, or transported feedback plus local presynaptic drive and voltage          |
| Dendritic coupling conductances                 | Child-to-parent tree transfer               | LocalCA conductance eligibility                                               | Same branch feedback field, child activity, parent voltage, and input resistance                                       |
| Reactivation slope/bias                         | Post-voltage transfer $f(V)$                | Configured local/autograd schedule                                            | Local activation derivative and available batch objective gradients; not part of the conductance-factorization theorem |
| Top- $k$ synapse banks                          | Sparse input selection and selected weights | Masked LocalCA on selected banks; inactive updates only when enabled          | Same branch feedback field, multiplied by top- $k$ masks and local modulators                                          |
| Linear decoder                                  | Readout from final core activity            | Local output-space decoder gradient or autograd                               | $\delta^y$ for local decoder mode; autograd for backprop mode                                                          |
| Nonlinear decoder                               | Compact CIFAR readout                       | Autograd for decoder parameters; Jacobian product for dendritic error mapping | $J_{\text{dec}}(h)^\top \delta^y$ maps output error into soma/core coordinates                                         |
| HSIC auxiliary                                  | Batch dependence regularizer                | Auxiliary gradient on selected local representations when configured          | Batch-global kernel gradient; reported as an auxiliary objective, not a local theorem term                             |
| Router / pathway roles                          | Structured cue-routing pathway selection    | Configuration-dependent router/pathway updates                                | Pathway-vector controls use router-derived role signals; this remains an exploratory higher-rank mechanism             |

Table S9: **Update scope by parameter family.** The exact factorization covers conductance-stage branch and dendritic-coupling parameters. Decoder, reactivation, HSIC, and router mechanisms are implementation-level training components and are therefore documented separately from the theorem.

## Morphology and Synapse Banks

The morphology notation  $[b_1, b_2, \dots, b_D]$  specifies a rooted dendritic tree. The factor  $b_d$  is the fan-in from dendritic stage  $d$  to the next more proximal stage. Thus  $[3, 3]$  gives 3 proximal branches per soma and 3 distal branches per proximal branch, for 9 distal leaves per soma. In the main feedforward models, synaptic inputs terminate on dendritic branches rather than directly on the soma. Unless a control explicitly changes this, each branch has  $N_E$  excitatory

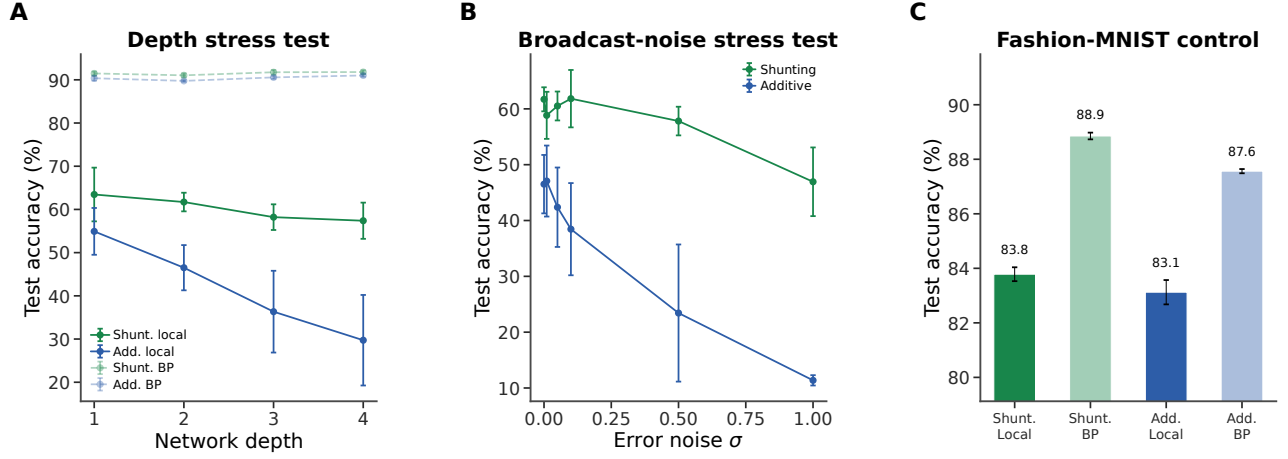


Figure S6: **Additional stress tests.** (A) Depth scaling: shunting local (green) degrades more gracefully from 63.5% to 57.4%; additive local (blue) drops from 54.9% to 29.7%. Matched backpropagation references (dashed) remain stable. (B) Broadcast-noise robustness: shunting remains stable for  $\sigma \leq 0.1$ ; additive degrades rapidly and reaches chance at  $\sigma = 1$ . (C) Fashion-MNIST: the shunting gain is modest on this cleaner benchmark (83.8% local vs. 88.9% backprop), consistent with the regime-dependent interpretation in the main text.

synapses and  $N_I$  inhibitory synapses. The branch input is sparse through a learned top- $k$  or banked synapse selection module; inactive synapses are masked out in the local update unless the corresponding control enables inactive-weight updates.

For a branch  $n$ , write  $x_{nj}^E, x_{nj}^I \geq 0$  for the selected excitatory and inhibitory presynaptic drives and  $a_c$  for the post-activation activity of child compartment  $c$ . The branch-level excitatory, inhibitory, and dendritic conductance currents are

$$U_n^E = \sum_{j=1}^{N_E} g_{nj}^E x_{nj}^E, \quad U_n^I = \sum_{j=1}^{N_I} g_{nj}^I x_{nj}^I, \quad D_n = \sum_{c \in \text{child}(n)} g_{c \rightarrow n}^{\text{den}} a_c.$$

The corresponding dendritic coupling load is

$$G_n^{\text{den}} = \sum_{c \in \text{child}(n)} g_{c \rightarrow n}^{\text{den}}.$$

For leaf branches,  $D_n = G_n^{\text{den}} = 0$ . For direct input-driven inhibition,  $x^I$  is produced by the inhibitory transfer stream. For explicit-I controls,  $x^I$  is the activity of the inhibitory-cell population.

## Shunting and Additive Forward Equations

In the shunting core, excitatory reversal is 1, leak and inhibitory reversal are 0, and leak conductance is fixed to 1. The implemented branch voltage is therefore

$$V_n^{\text{shunt}} = \frac{U_n^E + D_n}{1 + U_n^E + U_n^I + G_n^{\text{den}} + \varepsilon}.$$

The small numerical  $\varepsilon$  is used only for floating-point safety; analytically the leak conductance keeps the denominator positive. In implementation equations, the same stabilized denominator is used for the recorded input resistance:

$$g_n^{\text{tot}} = 1 + U_n^E + U_n^I + G_n^{\text{den}}, \quad g_{n,\varepsilon}^{\text{tot}} = g_n^{\text{tot}} + \varepsilon, \quad R_{n,\varepsilon}^{\text{tot}} = (g_{n,\varepsilon}^{\text{tot}})^{-1}.$$

The main analytical equations write  $R_n^{\text{tot}} = (g_n^{\text{tot}})^{-1}$ ; exact-gradient reconstruction tests use the implementation convention above. In the additive control, inhibition is subtractive rather than shunting. With the same nonnegative conductance variables and nonnegative inputs, the implemented comparator is

$$V_n^{\text{add}} = U_n^E + D_n - U_n^I,$$

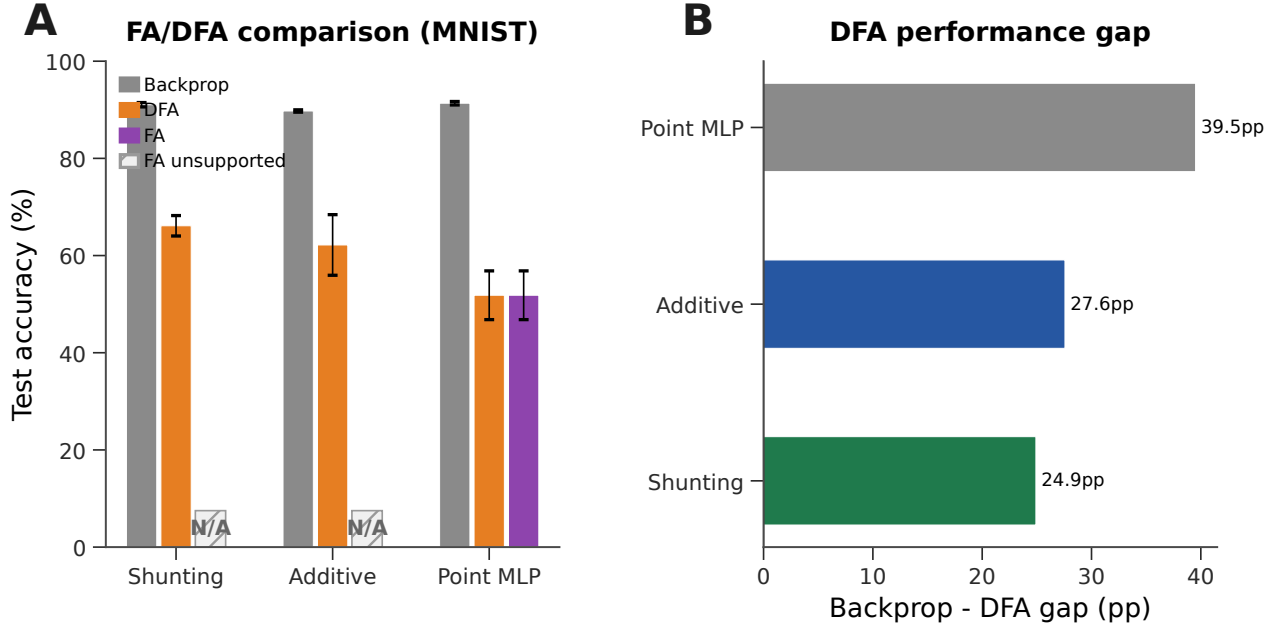


Figure S7: **Feedback alignment baselines (MNIST).** (A) Grouped comparison: standard backprop (neutral gray), DFA (orange), and FA (purple). Hatched N/A markers indicate that FA is not implemented for the block-structured dendritic architectures because the random feedback matrices are not dimensionally compatible with the tree-structured branch updates. DFA achieves 66.1% on shunting vs. 62.2% additive vs. 51.8% point MLP. (B) Backprop–DFA gap: dendritic architectures (24.9–27.6 pp) show smaller gaps than point MLPs (39.5 pp), suggesting conductance-based architecture is partially compatible with random feedback.

with optional additive-normalization controls reported separately in Fig. S9. Thus additive inhibition changes the signed voltage contribution and local additive eligibility, but it does not enter the denominator and does not gate conductance-stage path gain through  $R_n^{\text{tot}}$ .

### Positive Parameterization and Raw-Parameter Gradients

Trainable synaptic and dendritic conductances are stored as unconstrained raw parameters  $\theta$  and transformed by a positive map  $g = \psi(\theta)$ , with `softplus` used in the main reported runs:

$$\psi(\theta) = \log(1 + \exp \theta), \quad \psi'(\theta) = \sigma(\theta).$$

Equations in the main text give conductance-space gradients  $\partial L / \partial g$ . Before assigning gradients to the optimizer variables, the implementation applies the chain rule

$$\frac{\partial L}{\partial \theta} = \frac{\partial L}{\partial g} \psi'(\theta).$$

The exact-gradient reconstruction diagnostic includes this parameterization factor, as well as top- $k$  activity masks and dendritic block-linear parameterization. This is why the reconstruction comparison is against raw autograd gradients rather than only against abstract conductance variables.

### Post-Voltage Reactivation

The conductance stage produces a pre-reactivation voltage  $V_n$ , after which each branch may transmit a monotone post-voltage activity  $a_n = f_n(V_n)$ . The main sweeps use the learnable bounded transform

$$f_n(V) = \frac{\tanh(m_n(V - b_n)) + 1}{2}, \quad m_n = \exp(\ell_{m,n}),$$



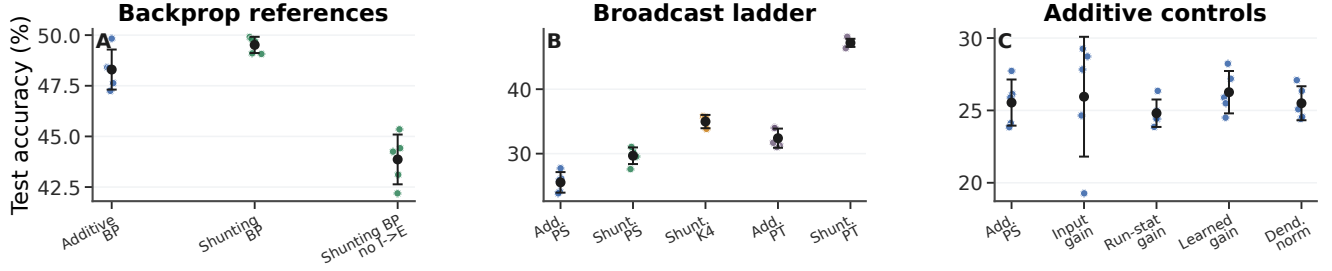


Figure S8: **CIFAR-10 control ladder in the compact direct-I-stream family.** (A) Matched backpropagation references. With nonnegative inputs and a nonlinear decoder, shunting slightly exceeds additive under standard training, and removing learned I-to-E conductance lowers the shunting reference. (B) Broadcast ladder. Per-soma feedback remains weak on this harder dataset, random low-rank  $K=4$  improves partially, and exact effective transported error nearly reaches the matched shunting backpropagation reference. (C) Additive fairness controls. Input-dependent gain, running-stat gain, learned gain, and dendritic normalization do not rescue additive per-soma feedback. This figure is a harder-dataset stress test, not a claim of competitive CIFAR-10 benchmarking.

with derivative

$$f'_n(V) = \frac{m_n}{2} [1 - \tanh^2(m_n(V - b_n))].$$

When reactivation is enabled, exact path transport uses the effective gain  $\tilde{\alpha}_n$ : parent-to-child transport includes the parent activation derivative, and the local update converts activation-space error into pre-activation voltage error by multiplying by the child derivative. With identity transfer, this reduces to conductance-stage transport through  $\alpha_n^{\text{cond}}$ .

## Decoder and Error Coordinates

Let  $h$  denote the final soma/core activity passed to the decoder and let  $\delta^y = \partial L / \partial \hat{y}$ . For a linear decoder  $\hat{y} = W_{\text{dec}} h$ , the soma/core activation-space teaching error is

$$\delta_0^a = W_{\text{dec}}^\top \delta^y.$$

For a nonlinear decoder, used in the CIFAR-10 compact control ladder, the implementation uses the decoder-input Jacobian product

$$\delta_0^a = J_{\text{dec}}(h)^\top \delta^y.$$

All LocalCA broadcast modes are defined in this soma/core activation-error coordinate system before local conversion to voltage-space errors. Decoder-update modes are: **backprop**, which uses autograd for decoder parameters; **local**, which is implemented for the linear readout and assigns

$$\hat{\nabla}_{W_{\text{dec}}} L = \langle \delta^y h^\top \rangle_B, \quad \hat{\nabla}_{b_{\text{dec}}} L = \langle \delta^y \rangle_B;$$

and **frozen**, which holds the decoder fixed. The local decoder gradient is output-space;  $\delta_0^a$  is the mapped soma/core activation-space error used by dendritic broadcasts.

## Biological Plausibility Assumptions

The model makes the following explicit assumptions. First, it uses the steady-state solution of the passive cable equation rather than temporal membrane dynamics; all path gains and gradient diagnostics refer to conductance-stage voltages at steady state. Second, each dendritic cell is a rooted tree, so each compartment has a unique path to the soma and the closed-form path gains  $\alpha_n^{\text{cond}}$  and  $\tilde{\alpha}_n$  are well-defined. Third, synaptic and dendritic conductance parameters are nonnegative by construction. Fourth, the main sweeps enforce nonnegative first-layer presynaptic drive through nonnegative inputs or a ReLU transfer stage. Fifth, post-voltage reactivation is handled through local

| Condition                                | Core     | Training / broadcast | I-to-E | Test acc.         |
|------------------------------------------|----------|----------------------|--------|-------------------|
| standard                                 | additive | backprop             | yes    | 48.3 ± 1.0        |
| standard                                 | shunting | backprop             | yes    | <b>49.5 ± 0.4</b> |
| standard, no learned inhibition          | shunting | backprop             | no     | 43.9 ± 1.2        |
| per-soma                                 | additive | 5F per-soma          | yes    | 25.5 ± 1.6        |
| per-soma, input-dependent gain           | additive | 5F per-soma          | yes    | 25.9 ± 4.1        |
| per-soma, running-stat gain              | additive | 5F per-soma          | yes    | 24.8 ± 0.9        |
| per-soma, learned gain                   | additive | 5F per-soma          | yes    | 26.3 ± 1.5        |
| per-soma, dendritic normalization        | additive | 5F per-soma          | yes    | 25.5 ± 1.2        |
| per-soma                                 | shunting | 5F per-soma          | yes    | 29.7 ± 1.3        |
| per-soma, no learned inhibition          | shunting | 5F per-soma          | no     | 22.3 ± 0.7        |
| transported error                        | additive | 5F path transport    | yes    | 32.4 ± 1.5        |
| rank-4                                   | shunting | 5F low-rank          | yes    | 35.0 ± 1.0        |
| transported error                        | shunting | 5F path transport    | yes    | <b>47.2 ± 0.6</b> |
| transported error, no learned inhibition | shunting | 5F path transport    | no     | 38.6 ± 1.0        |

Table S4: **Exact CIFAR-10 values for the 70-run control ladder (5 seeds per condition, validation-best epoch).** All rows use flattened CIFAR-10 in  $[0, 1]$ , nonnegative transfer outputs, positive conductance transforms, a compact depth-4 dendritic core, and a nonlinear decoder. LocalCA maps output error back to decoder-input/soma coordinates through the decoder Jacobian. The useful conclusions are narrow but important: learned shunting slightly improves the matched backpropagation reference, learned I-to-E conductance is necessary for the strongest shunting results, transported error nearly closes the shunting LocalCA gap, and additive per-soma fairness controls do not explain the transported-shunting result.

| Inhibitory input mode    | Training / broadcast                             | Test accuracy        |
|--------------------------|--------------------------------------------------|----------------------|
| Direct inhibitory stream | LocalCA, per-soma feedback                       | 0.852 ± 0.007        |
| Direct inhibitory stream | LocalCA, exact effective path transport          | <b>0.952 ± 0.000</b> |
| Explicit I cells         | LocalCA, path transport, train I-cell dendrites  | 0.942 ± 0.002        |
| Explicit I cells         | LocalCA, path transport, freeze I-cell dendrites | 0.938 ± 0.002        |
| Explicit I cells         | standard backprop                                | 0.951 ± 0.003        |

Table S5: **One-feedforward-layer input-mode path-gain probe on noise resilience (3 seeds).** All rows use one excitatory dendritic population with  $[3, 3]$  morphology and nonnegative inputs. The direct-stream rows have no explicit inhibitory neurons; the input stream drives learned branch-level inhibitory conductance banks directly. The explicit-I rows use a matched inhibitory population with  $[3, 3]$  morphology. Exact effective path transport closes the direct-stream per-soma gap, and explicit inhibitory cells nearly match both direct-stream path transport and the explicit-I matched backpropagation reference.

$f'(V)$  factors along the path. Sixth, each dendritic synapse receives a restricted non-local feedback field—scalar, per-soma, low-rank, path-structured, or transported-oracle depending on the condition—while presynaptic activity, voltage, reversal potential, input resistance, and slowly estimated branch-level modulators are local or branch-local quantities.

## Units and Parameterization

Table S10 summarizes the numerical conventions used throughout the implementation and diagnostics, including which quantities are constrained conductances and which are unconstrained readout parameters.

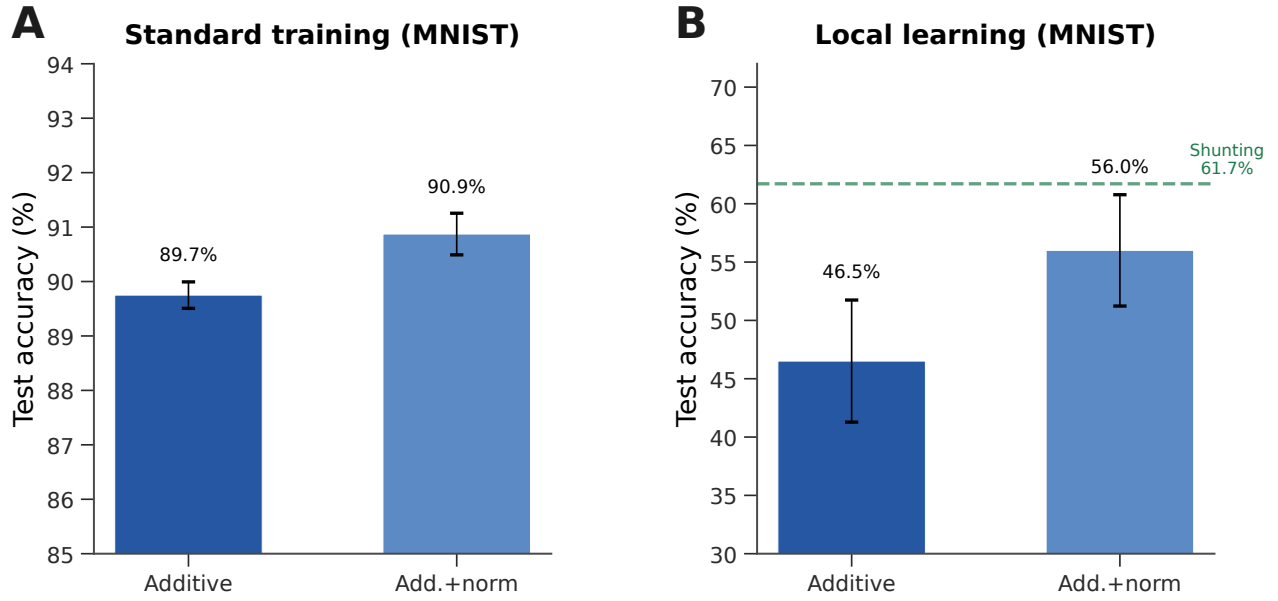


Figure S9: **Additive + normalization control (MNIST stress-regime diagnostic).** (A) Standard backprop: normalization provides a small boost (89.7%  $\rightarrow$  90.9%). (B) Local learning: normalization improves additive from 46.5% to 56.0% (+9.5 pp), partially closing the gap to shunting (61.7%, dashed green). These lower local accuracies come from a separate stress-regime diagnostic, not from the matched-capacity performance setting in Table S3.

## Representative Manuscript Architectures

| Setting                                                    | Encoder  | E layers           | Tree                                               | Synapses / branch                                             | Seeds                   | Notes                                                                                                                                              |
|------------------------------------------------------------|----------|--------------------|----------------------------------------------------|---------------------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Gradient fidelity / path-gain / noise-resilience mechanism | id.+ReLU | [128, 128]         | [3, 3]                                             | $N_E=40$ ,<br>$N_I \in \{0, 5, 10, 20, 40\}$                  | 5 fidelity,<br>5 oracle | Main mechanism sweeps in Figs. 2 and 3; reactivation <code>param_tanh</code> .                                                                     |
| MNIST / figure-ground local competence                     | id.+ReLU | [128]              | [3, 3]                                             | $N_E=40$ , $N_I=20$                                           | 5                       | Best local 5F per-soma runs in Table S3; reactivation <code>param_tanh</code> .                                                                    |
| Fashion-MNIST competence                                   | id.+ReLU | [128]              | [3, 3]                                             | $N_E=40$ , $N_I=20$                                           | 5                       | Dedicated five-seed standard-vs-local competence sweep; reactivation <code>param_tanh</code> .                                                     |
| Cue-routing PV-LocalCA                                     | router   | [64]               | [2]                                                | $N_E=10$ , $N_I=4$                                            | 5                       | Learned router with two pathway groups and rank-2 structured feedback; reactivation <code>param_tanh</code> .                                      |
| CIFAR-10 compact direct-I-stream control ladder            | id.+ReLU | [20] E, no I cells | [3, 3, 3, 3]                                       | $N_E=25$ , $N_I=25$ direct I-to-E, with matched no-I controls | 5 per condition         | Harder-dataset stress test with input-driven inhibitory conductance, decoder [32, 16], decoder-aware soma mapping, and additive fairness controls. |
| Morphology $\times$ inhibition appendix map                | id.+ReLU | [128, 128]         | [2, 2], [3, 3],<br>[4, 4], [2, 2, 2],<br>[3, 3, 3] | $N_E=40$ ,<br>$N_I \in \{0, 5, 10, 20, 40\}$                  | 3                       | Supportive 3-seed sweep; reactivation <code>param_tanh</code> .                                                                                    |

Table S11: **Representative architectures used in the manuscript.** All dendritic rows use explicit branch-level excitatory and inhibitory synapse banks without direct somatic synapses, nonnegative conductances, and a nonnegative first-layer drive enforced by an explicit ReLU transfer stage in the main runs. The tree column gives dendritic morphology per excitatory neuron, and the synapse counts report excitatory and inhibitory synapses per branch.

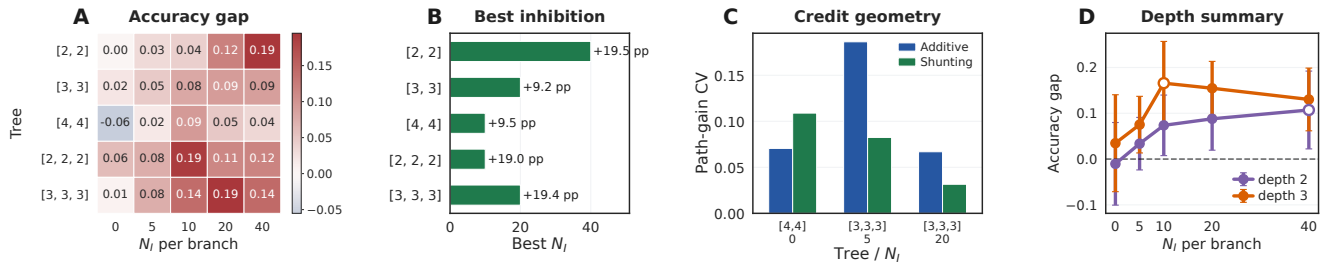


Figure S10: **Morphology shifts the useful inhibitory regime.** (A) Shunting-minus-additive LocalCA accuracy gap across dendritic morphologies and inhibitory synapse counts. (B) Best inhibitory count for each morphology. (C) Representative path-gain diagnostic for selected morphology cells, used to connect the performance map to credit geometry. (D) Average shunting advantage by depth, with markers showing the peak inhibitory count for each depth.

## Hyperparameters

Table S12 collects the core training hyperparameters for each main experiment. Learning rates are applied through parameter groups (top- $k$ , dendritic block-linear, reactivation, decoder) with the values given in the representative config files; where a single LR is reported, every group uses that value.

| Setting                                                                  | Opt. | LR                                             | Batch | Epochs | Weight decay | Notes                                                     |
|--------------------------------------------------------------------------|------|------------------------------------------------|-------|--------|--------------|-----------------------------------------------------------|
| MNIST / Fashion-MNIST / figure-ground MNIST (5F per-soma LocalCA)        | Adam | $10^{-3}$ / $5 \cdot 10^{-4}$ (block / react.) | 256   | 100    | 0            | fixed-epoch, no early stopping; param_tanh reactivation   |
| MNIST / Fashion-MNIST / figure-ground MNIST (matched backprop reference) | Adam | $10^{-3}$ / $5 \cdot 10^{-4}$                  | 256   | 100    | 0            | matched to LocalCA setup                                  |
| Gradient fidelity / path-gain / noise resilience                         | Adam | $10^{-3}$                                      | 256   | 50     | 0            | 5 seeds, hooks enabled for diagnostic capture             |
| Morphology $\times$ inhibition regime map (supportive)                   | Adam | $10^{-3}$                                      | 256   | 50     | 0            | 3 seeds                                                   |
| Cue routing                                                              | Adam | $10^{-3}$                                      | 256   | 90     | 0            | early stopping with patience 30                           |
| CIFAR-10 compact direct-I-stream depth-4 LocalCA                         | Adam | $10^{-3}$ / $10^{-4}$ (block / react.)         | 256   | 400    | 0 / 0.01     | grad-clip 5.0, early stop patience 50, nonlinear decoder  |
| CIFAR-10 compact direct-I-stream depth-4 BP                              | Adam | $10^{-3}$                                      | 256   | 200    | 0.01         | early stop patience 40, matched backpropagation reference |

Table S12: **Training hyperparameters by experiment.** All runs use Adam with default  $\beta_1=0.9$ ,  $\beta_2=0.999$ ,  $\epsilon=10^{-8}$ . Reactivation and block-linear parameter groups use a slower LR to keep conductance-level dynamics stable. Representative configs are provided in the accompanying repository.

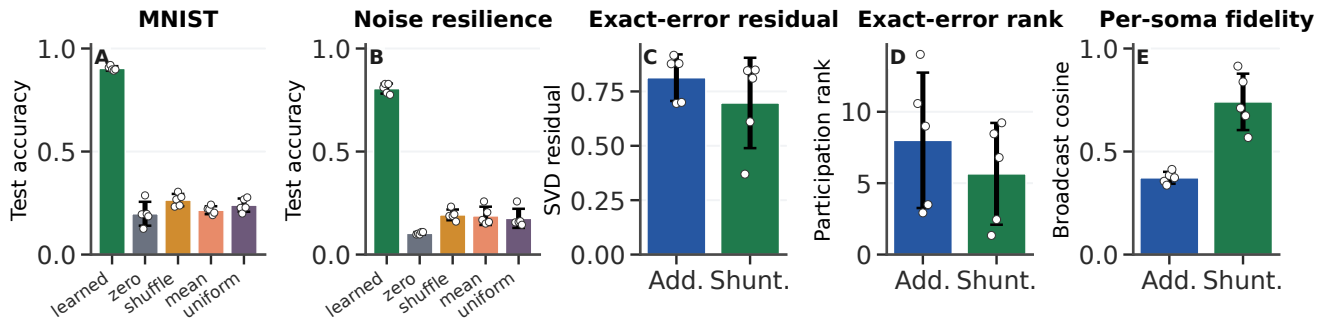


Figure S11: **Post-training inhibition intervention and exact-error compressibility.** (A,B) Post-training inhibitory-conductance interventions on trained shunting LocalCA checkpoints. Accuracy is measured on a 512-example held-out subset; dots show seeds. Removing, shuffling, or clamping learned inhibition collapses performance, showing that the learned inhibitory field is sample-dependent and functionally important. (C,D) Exact-error diagnostics on matched MNIST  $N_I=5$  runs. Shunting has a smaller unrestricted rank-1 residual, lower participation rank, and higher actual per-soma broadcast cosine than additive, so the exact error field is both more intrinsically compressible and better aligned with the implemented feedback field. Error bars are s.d. across 5 seeds.

### Effective Broadcast Dimensionality in Main Architectures

| Setting                                          | Soma/core dim. | Output classes | Main feedback field                                                                                           |
|--------------------------------------------------|----------------|----------------|---------------------------------------------------------------------------------------------------------------|
| Gradient fidelity / path-gain / noise resilience | 128, 128       | 10             | Per-soma soma/core error at matched-width dendritic layers                                                    |
| MNIST / Fashion-MNIST competence                 | 128            | 10             | Per-soma soma/core error                                                                                      |
| Figure-ground MNIST competence                   | 128            | 10             | Per-soma soma/core error                                                                                      |
| Cue routing                                      | 64             | 2              | Tuned per-soma baseline; higher-rank tests use low-rank or pathway-vector feedback                            |
| Compact CIFAR-10 harder-dataset stress test      | 20             | 10             | Vector-valued per-soma after decoder-aware mapping; higher-rank tests use explicit low-rank or path transport |

Table S13: **Effective dimensionality of the main feedback in the reported architectures.** Output loss gradients are first mapped into decoder-input/soma coordinates. The main competence and mechanism runs use the implemented per\_soma mode when the soma/core dimension matches layer width; scalar, low-rank, pathway-vector, and transported-error conditions are labeled separately as controls or oracles.

### Compute Resources

All experiments were run on an institutional GPU cluster. Reproduction runs use one NVIDIA A100 or H100-class GPU (40–80GB memory) per seed and do not require distributed training. Table S14 gives approximate single-GPU runtime ranges for the reported experiment families; unreported exploratory runs used additional cluster time and are not counted in the reproduction estimate.

| Model implication                                                                                                                                               | Biological prediction                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Path-local inhibitory conductance changes $\alpha_n^{\text{cond}}$ only for descendants whose credit path crosses the inhibited compartment.                    | Branch-local inhibitory perturbation should distort plasticity or teaching-signal efficacy for synapses below the perturbed compartment more than for sister branches. |
| The log derivative $\partial \log \alpha_n^{\text{cond}} / \partial G_k^I = -R_k^{\text{tot}}$ predicts stronger credit gating in high-resistance compartments. | The same inhibitory conductance change should have larger effects on low-conductance/high-resistance branches than on already high-conductance branches.               |
| Single-channel broadcast succeeds when exact compartment errors are effectively shared across the relevant branches and fails when branch identity matters.     | Tasks requiring route- or context-specific branch credit should need richer dendritic teaching signals than a single scalar modulatory broadcast.                      |

Table S6: **Experimental predictions from the path-gain mechanism.** These are qualitative biological predictions, not claims tested directly in this paper.

| Setting                                                           | Broadcast                      | Test accuracy                       |
|-------------------------------------------------------------------|--------------------------------|-------------------------------------|
| <i>Noise resilience, shunting, rank bridge</i>                    |                                |                                     |
| per-soma                                                          | per-soma feedback              | $0.662 \pm 0.016$                   |
| path propagation                                                  | recursive attenuation proxy    | $0.762 \pm 0.047$                   |
| low-rank                                                          | $K=1$                          | $0.453 \pm 0.126$                   |
| low-rank                                                          | $K=2$                          | $0.630 \pm 0.042$                   |
| low-rank                                                          | $K=4$                          | $0.764 \pm 0.025$                   |
| low-rank                                                          | $K=8$                          | $0.795 \pm 0.019$                   |
| path transport                                                    | exact effective path transport | <b><math>0.847 \pm 0.018</math></b> |
| <i>Cue routing, learned shunting router, rank/structure sweep</i> |                                |                                     |
| per-soma                                                          | tuned per-soma feedback        | $0.803 \pm 0.125$                   |
| low-rank                                                          | $K=1$                          | $0.742 \pm 0.022$                   |
| low-rank                                                          | $K=2$                          | <b><math>0.846 \pm 0.082</math></b> |
| low-rank                                                          | $K=4$                          | $0.748 \pm 0.085$                   |
| pathway-vector                                                    | tuned structured rank-2        | $0.815 \pm 0.126$                   |

Table S7: **Rank bridge and rank-structure comparison.** On shunting noise resilience, both recursive path propagation and higher-rank random low-rank broadcast close a substantial fraction of the gap between per-soma feedback and exact effective path transport, with low-rank continuing to improve up to  $K=8$ . On cue routing, the learned-router sweep shows that moving beyond the per-soma baseline helps, but the best unstructured low-rank setting ( $K=2$ ) is still slightly better than the tuned structured pathway-vector variant. We therefore interpret routed credit assignment as an open higher-rank problem rather than as a solved pathway-vector rescue.

| Experiment family                                       | Runs       | Time / run  | Approx. GPU-h |
|---------------------------------------------------------|------------|-------------|---------------|
| Performance and verification                            | $\sim 25$  | 5–20 min    | 4–9           |
| Gradient, path-gain, inhibition, and oracle diagnostics | $\sim 80$  | 10–30 min   | 15–40         |
| Morphology, stress, rule, and feedback controls         | $\sim 120$ | 5–35 min    | 20–70         |
| CIFAR-10 control ladder                                 | 70         | 0.8–2.5 h   | 60–175        |
| Figure generation and CPU-side summaries                | –          | < 5 h total | < 5           |

Table S14: **Approximate compute for reported results.** Ranges reflect single-seed wall-clock variation across A100/H100-class GPUs, data-loading overhead, and early stopping.

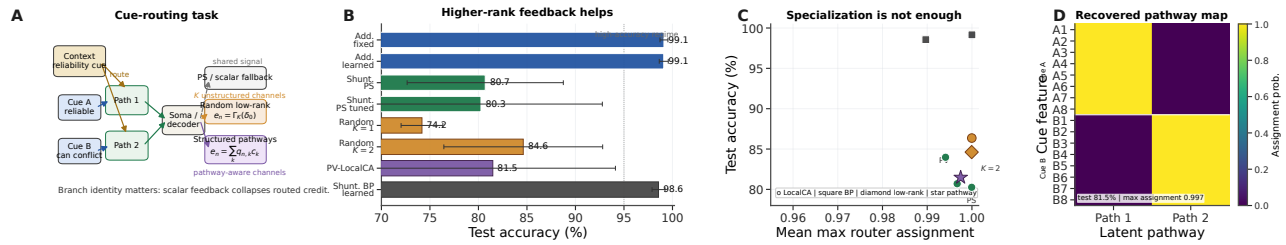


Figure S12: **Structured feedback when restricted feedback fails.** (A) Cue-routing task schematic. Context determines which noisy cue is reliable on each trial, so different dendritic pathways should receive different teaching signals. (B) Five-seed comparison of per-soma, random low-rank, and pathway-structured feedback. Moving beyond the per-soma / single-channel baselines helps, while the best structured repair remains open. (C) Test accuracy vs. router specialization across learned-router conditions. High specialization alone does not guarantee success. (D) Recovered pathway assignments for a representative learned-router run. Cue-A and cue-B features segregate into separate latent pathways even when feedback quality, rather than router discreteness, is the limiting factor.

## Code, Data, and Asset Availability

MNIST, Fashion-MNIST, and CIFAR-10 are public datasets or public benchmark assets cited in the paper; we use them under their standard published access conditions and cite their original sources in the bibliography. The accompanying repository contains the core model and training code, selected diagnostic utilities, representative configuration files, tests, and reproduction instructions for the main experimental families. A public release should identify the immutable commit or release tag used for the posted manuscript. The paper does not release a new dataset or a stand-alone pretrained model asset. Dataset access pages are the original MNIST site / UCI entry (<https://archive.ics.uci.edu/dataset/683/mnist+database+of+handwritten+digits>), the Fashion-MNIST repository (<https://github.com/zalandoresearch/fashion-mnist>), and the CIFAR page (<https://www.cs.toronto.edu/~kriz/cifar.html>).

| Asset           | Use in this paper                                            | Access / license or terms note                                                                                                                                                     |
|-----------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MNIST           | Digit classification and derived nonnegative synthetic tasks | Public benchmark from the original MNIST site / UCI entry; UCI lists DOI 10.24432/C53K8Q and asks users to follow the original acknowledgement policy; no raw-data redistribution. |
| Fashion-MNIST   | Apparel classification stress test                           | Public Zalando Research benchmark; repository is MIT licensed; no raw-data redistribution beyond standard dataset loaders.                                                         |
| CIFAR-10        | Flattened harder-dataset stress test                         | Public CIFAR dataset site asks users to cite Krizhevsky’s technical report; no raw-data redistribution.                                                                            |
| Synthetic tasks | Figure-ground MNIST, noise resilience, cue integration       | Generated procedurally from public benchmarks or random seeds described in Appendix D; no new third-party asset is introduced.                                                     |

Table S15: **Existing assets used in the manuscript.** We credit the original sources in the bibliography, use public benchmark assets under their published access conditions, and do not redistribute third-party raw data in the manuscript package.

| Quantity         | Symbol                           | Convention                                                                                                             |
|------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Voltage          | $V$                              | Conductance-stage shunting voltage in $[0, 1]$ under nonnegative drive; additive controls may be signed                |
| Conductances     | $g^{\text{syn}}, g^{\text{den}}$ | Nonnegative via $g = \text{softplus}(\theta)$ ; local gradients are mapped to raw $\theta$ by the transform derivative |
| Leak conductance | $g^{\text{leak}}$                | Set to 1                                                                                                               |
| Input resistance | $R^{\text{tot}}$                 | $\leq 1$                                                                                                               |

Table S10: Units and normalization.

## LocalCA Broadcast and Gradient Assignment

LocalCA first maps the output loss derivative to the soma/core activation-space error  $\delta_0^a$  and then constructs a branch voltage-error field. In the final dendritic layer this is the decoder-input error; in earlier matched-width layers the practical implementation reuses the same coordinate by neuron index as an approximate layer-soma teaching signal. This is an inter-layer feedback approximation, separate from the within-tree path-transport approximation studied by the main theory. In the main reported `per_soma` mode, if the layer width matches the soma/core error dimension, each soma-specific error component is broadcast to the corresponding dendritic compartments:

$$e_{n,b}^a = \delta_{0,b}^a \in \mathbb{R}^{d_0}.$$

The stricter scalar control compresses the soma/core error to one value per example,

$$\bar{\delta}_b = \frac{1}{d_0} \sum_{c=1}^{d_0} \delta_{0,bc}^a, \quad e_{n,b}^a = \bar{\delta}_b \mathbf{1}_n,$$

where  $d_0$  is the dimension of the soma/core error. If dimensions do not match, the implementation falls back to scalar-style expansion rather than silently inventing an unmatched per-soma map. The low-rank controls use fixed random projection and mixing matrices  $P_K$  and  $Q_n$ ,

$$c_b = P_K \delta_{0,b}^a \in \mathbb{R}^K, \quad e_{n,b}^a = Q_n c_b.$$

The pathway-vector controls use router-inferred pathway roles to gate and transport a low-rank role vector. The transported oracle uses the effective tree recursion. If  $p$  is the parent of child  $c$ , activation-space error is propagated by

$$e_c^a = e_p^a f'_p(V_p) R_{p,\varepsilon}^{\text{tot}} g_{c \rightarrow p}^{\text{den}},$$

and the local pre-reactivation voltage error used in conductance eligibility is

$$e_n^V = \begin{cases} e_n^a f'_n(V_n), & \text{post-voltage reactivation enabled,} \\ e_n^a, & \text{identity transfer.} \end{cases}$$

This operational definition is the implementation counterpart of the effective path gain  $\tilde{\alpha}_n$  in Theorem 1.

For shunting branches, the conductance-space LocalCA gradients assigned before raw-parameter transformation are

$$\widehat{\nabla}_{g_{nj}^E} L = \langle e_n^V x_{nj}^E R_{n,\varepsilon}^{\text{tot}} (1 - V_n) \rangle_B, \quad \widehat{\nabla}_{g_{nj}^I} L = \langle e_n^V x_{nj}^I R_{n,\varepsilon}^{\text{tot}} (0 - V_n) \rangle_B,$$

and for a dendritic coupling from child  $c$  to parent  $n$ ,

$$\widehat{\nabla}_{g_{c \rightarrow n}^{\text{den}}} L = \langle e_n^V R_{n,\varepsilon}^{\text{tot}} (a_c - V_n) \rangle_B.$$

For additive controls, the matching local derivatives are

$$\widehat{\nabla}_{g_{nj}^E} L = \langle e_n^V x_{nj}^E \rangle_B, \quad \widehat{\nabla}_{g_{nj}^I} L = -\langle e_n^V x_{nj}^I \rangle_B, \quad \widehat{\nabla}_{g_{c \rightarrow n}^{\text{den}}} L = \langle e_n^V a_c \rangle_B.$$

The optimizer applies the usual descent step; equivalently, the sign can be absorbed into the definition of the broadcast error. After these conductance-space gradients are formed, they are multiplied by the positive-transform derivative  $\psi'(\theta)$  and by any active synapse masks before assignment to raw parameters.



## 4F and 5F Local Modulators

The 3F rule uses only the local eligibility and broadcast error above. The 4F and 5F variants multiply the conductance-space gradient by branch-level reliability factors before raw-parameter transformation. For 4F, the implementation records  $V_n$  from each branch layer and the corresponding soma/core activity  $V_0$ . For each batch item  $b$ , it first averages over the non-batch coordinates of the recorded layer tensor to obtain scalar summaries  $\bar{V}_{n,b}$  and  $\bar{V}_{0,b}$ . It then estimates covariance over the batch axis and smooths the resulting scalar with an EMA:

$$r_n^{4F} = \frac{\text{Cov}_B(\bar{V}_{n,b}, \bar{V}_{0,b})}{\sqrt{\text{Var}_B(\bar{V}_{n,b})\text{Var}_B(\bar{V}_{0,b})} + \varepsilon}.$$

The EMA is initialized from the first observed batch statistic; the recorded tensors are detached from autograd, and the same mini-batch supplies both the statistics and the gradient update. This proxy is clamped to the positive stability range  $[0.1, 2.0]$  before it multiplies the local gradient; sign-opposed covariance therefore reduces the multiplier rather than reversing update direction. Because the smoothed numerator and denominator can be updated separately, and because single-sample fallbacks use online covariance moments, the clamped value can exceed one and should be read as a preconditioner rather than a literal Pearson correlation. The 5F factor adds a bounded branch-level preconditioner based on how much branch voltage variance remains predictable from parent or soma-level activity:

$$\phi_n = \text{clip}_{[0.25, 4.0]} \left( \frac{\text{Var}(V_n)}{\sigma_{\text{res},n}^2 + \varepsilon} \right),$$

where  $\sigma_{\text{res},n}^2$  is estimated online from a scalar ridge regression of the branch-layer voltage on the configured parent proxy. Batch-mode estimates center over examples and flatten remaining layer coordinates before updating the smoothed variance and covariance moments; single-sample fallback uses scalar online moments. Since these moments are smoothed and clamped rather than recomputed as a single ordinary least-squares fit on a fixed batch,  $\phi_n$  can transiently fall below one or rise above one. The lower clamp keeps noisy residual estimates from suppressing a branch-layer update entirely, and the upper clamp prevents highly predictable branches from dominating optimization. In the main 5F runs, the multiplier is  $r_n^{4F} \phi_n$ . The 5F sensitivity diagnostic in Fig. S2 varies the clamp and EMA rate to verify that the reported MNIST performance is not a single brittle clamp setting.

## Algorithm

Algorithm 1 gives the update order for the main 5F per-soma LocalCA condition and shows where scalar, higher-bandwidth, and transported-error controls enter.

---

### Algorithm 1 Main 5F Per-Soma LocalCA Update

---

- 1: **Input:** Dendritic model, batch  $(x, y)$ , optimizer
  - 2: Forward pass; loss  $L$ , output error  $\delta^y$
  - 3: Somatic/core activation error  $\delta_0^a = \text{decoder-input Jacobian product } J_{\text{dec}}(h_{\text{core}})^\top \delta^y$  (or  $W_{\text{dec}}^\top \delta^y$  for a linear decoder)
  - 4: **for** each layer  $n$  (reverse) **do**
  - 5:   Per-soma feedback  $e_n^a = \delta_0^a$  when dimensions match; scalar/low-rank/path controls replace this line
  - 6:   Update EMA estimates for  $r_n^{4F}$  and  $\phi_n$  from batch voltages
  - 7:   Convert to pre-activation voltage error  $e_n^V$  using the local activation derivative when needed
  - 8:    $\hat{V}_{g_j^{\text{syn}}} L \leftarrow r_n^{4F} \phi_n \langle x_j R_{n,\varepsilon}^{\text{tot}} (E_j - V_n) e_n^V \rangle_B$
  - 9:    $\hat{V}_{g_c^{\text{den}}} L \leftarrow r_n^{4F} \phi_n \langle R_{n,\varepsilon}^{\text{tot}} (a_c - V_n) e_n^V \rangle_B$
  - 10:   Map conductance gradients to raw parameters by multiplying by  $\partial g / \partial \theta$
  - 11: **end for**
  - 12: Optional controls replace line 5 with scalar, rank- $K$ , path-structured, or transported-oracle feedback.
  - 13: Clip gradients; optimizer step
-

## C Theoretical Details

This section collects theory-adjacent material that supports implementation and interpretation but is not part of the main proof. The central derivation remains Theorem 1, Corollary 1, and Prop. 2.

### Random-Broadcast Alignment Intuition

Feedback-alignment arguments suggest that random or low-rank feedback can be useful when the induced local update remains positively correlated with the exact gradient [14]. In this model, that correlation depends on how local eligibility factors co-vary with the conductance-stage path gain in Eq. (4). We treat this as intuition only. The central theoretical object in the paper is still the exact pre-reactivation compartment error  $\partial L / \partial V_n$ ; under identity upward transfer this reduces to  $\alpha_n^{\text{cond}} \delta_0^V$ , while post-voltage activation uses  $\tilde{\alpha}_n \delta_0^V$ . The empirical question is how faithfully different broadcast fields approximate that quantity.

### Morphology-Aware Extensions

The following variants are implemented controls or architectural extensions used to test whether coarse morphology-aware feedback can approximate exact path transport. They are not required for the main 5F per-soma results, but they define the path-propagation, depth, normalization, and pathway-vector controls reported in the appendix figures and tables.

**Path-integrated propagation.** Modulate broadcast error by  $\pi_n = \pi_{n-1} \cdot R_{n-1}^{\text{tot}} \cdot \bar{g}_{n-1}^{\text{den}}$ , approximating depth attenuation from Eq. (4) without computing the exact sample-specific path gain.

**Depth modulation.** Per-branch scaling  $\kappa_j = \kappa_{\text{base}} / (d_j + d_0)$ , mirroring cable attenuation and testing whether a simple depth prior can stabilize distal updates.

**Dendritic normalization.**  $\Delta g_j^{\text{den}} \leftarrow \Delta g_j^{\text{den}} / (\sum_k g_k^{\text{den}} + \epsilon)$ , analogous to homeostatic scaling [26] and used as an additive-control normalization comparison.

**Pathway-vector feedback.** For tasks with latent pathway structure, the broadcast becomes a role vector inferred from the router. Local pathway activity gates this vector. Upward block transport then passes it to earlier branches, aligning their feedback with downstream descendants.

**Router-derived branch roles.** Each branch receives a role profile inferred from router assignments and incoming block weights rather than from a hand-coded apical/basal label. Plasticity is then modulated by branch selectivity and a synapse-specific role-alignment factor, emphasizing pathway-consistent updates without imposing a heuristic branch taxonomy.

### HSIC Auxiliary Objectives

Following [24], the figure-ground MNIST LocalCA runs optionally apply an HSIC-style auxiliary gradient to selected layer representations  $\mathbf{Z}$ :

$$\mathcal{L}^{\text{self}} = B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Z \mathbf{H}), \quad \mathcal{L}^{\text{target}} = -B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Y \mathbf{H}).$$

Here  $B$  is batch size,  $\mathbf{H}$  is the centering matrix,  $\mathbf{K}_Z$  uses the configured representation kernel, and  $\mathbf{K}_Y$  uses one-hot class labels when `target_source=labels`. The reported figure-ground runs use an RBF kernel with fixed bandwidth  $\sigma = 1.0$ , HSIC weight 0.01, self and target weights 0.3, five-epoch warmup, and gradient clipping at 0.1; the code applies the resulting auxiliary gradient through the configured local-learning representation pathway rather than through the conductance-factorization theorem. Moderate weights (0.01–0.1) help on figure-ground MNIST; negligible on MNIST. Online statistics ( $r_n^{\text{4F}}$ ,  $\phi_n$ ) use Welford’s algorithm [25].

## D Task Construction and Robustness Protocols

This section specifies the benchmark, synthetic-task, depth, and feedback-noise protocols used in the experiments, with emphasis on which inputs preserve the nonnegative-drive regime assumed by the conductance model.

### General Data Handling

MNIST, Fashion-MNIST, and CIFAR-10 are loaded from standard public dataset loaders, flattened before entering the dendritic core, scaled to  $[0, 1]$ , and passed through the configured nonnegative transfer stage in the main experiments. MNIST-derived synthetic tasks preserve the flattened  $28 \times 28$  geometry. Fixed random seeds define task generators such as corruption projections and context masks; generator parameters are shared across train, validation, and test splits, while sample-level corruptions are drawn per example. Validation accuracy selects checkpoints, and test accuracy is reported. CIFAR-10 is used as a flattened harder-dataset stress test under the same conductance constraints rather than as a competitive vision benchmark; no data augmentation is used as a performance device.

### Supervised Benchmark Tasks

**MNIST and Fashion-MNIST.** These ten-class tasks use nonnegative pixel inputs. MNIST tests whether per-soma 5F LocalCA approaches matched backpropagation on clean supervised data; Fashion-MNIST adds a same-dimensional distribution shift.

**CIFAR-10 harder-dataset stress test.** CIFAR-10 inputs are flattened RGB images in  $[0, 1]$ . The compact CIFAR family uses a smaller dendritic core, nonlinear decoder, and decoder-aware LocalCA mapping to test whether the broadcast-fidelity ladder persists on harder data: per-soma feedback is weak, higher-rank feedback helps, and exact effective path transport approaches the matched shunting backpropagation reference.

**Figure-ground MNIST.** For each reshaped  $28 \times 28$  MNIST image, the right half carries the digit and the left half is replaced by independent distractors  $u_{r,c} \sim \text{Uniform}(0, 0.25)$ , clipped to  $[0, 1]$  when needed. The task tests whether local credit assignment exploits spatial signal/distractor structure without leaving the nonnegative-drive regime. The main LocalCA competence runs use the HSIC auxiliary objective with weight 0.01, which adds roughly 3 pp in the ablation.

**Noise resilience.** Flattened MNIST images are corrupted as  $\tilde{x} = \text{clip}(x + \sigma_{\text{task}}Az, 0, 1)$  with  $\sigma_{\text{task}} = 1.5$ , a fixed task-seeded projection  $A \in \mathbb{R}^{784 \times 50}$ , and fresh  $z \sim \mathcal{N}(0, I_{50})$  for each corrupted example. The fixed projection creates correlated interference without reducing the task to memorizing one corruption pattern.

**Cue integration.** Two noisy cue streams ( $A$  and  $B$ ) are presented simultaneously for two-class classification, with a one-hot context indicating which cue is reliable on each trial. Cue vectors are clipped to  $[0, 1]$  after noise. The fixed-pathway variant duplicates context into both cue branches; the learned-routing variant requires the router to discover cue separation from data. This feedback-rank diagnostic tests structured pathway-vector broadcast inside the nonnegative-input regime.

### Depth Scaling and Noise Robustness

These stress diagnostics keep the same broad model family while changing dendritic depth or corrupting the broadcast signal. For depth scaling, dendritic depth varies from 1–4 layers (branch factors [9] to [3, 3, 3, 3]): shunting local degrades from 63.5% to 57.4%, additive local falls from 54.9% to 29.7%, the shunting advantage grows from +8.5 to +27.7 pp, and matched backpropagation references remain near 90–92% (Fig. S6A). For noise robustness, Gaussian noise  $\mathcal{N}(0, \sigma^2)$  is added to the broadcast error; shunting remains robust to  $\sigma \leq 0.1$  (about 62%), while additive drops from 46.5% to chance at  $\sigma=1.0$ , indicating that shunting credit signals carry useful learning information beyond broadcast magnitude alone (Fig. S6B).